Pretraining evolution across the campaign generations (v4, v5, v6)

This document records how the pretraining stage of the dfs6311 campaigns changed from the v4 generation (unanchored point-wise fits to the parent's enhancement factors), through v5 (per-rung self-consistent seeding around an unchanged pretraining scheme), to v6 (parent-anchored networks with a machine-checked fidelity certificate), together with the measured findings that forced each change and the first quantitative comparison of the completed v6 GGA cells against the corrected v4 GGA record. Every number carries its source: an entry of `xcquinox/alec/HISTORY.md` (cited as `HISTORY <date>`), a code definition (cited as `file:symbol` or `file:line`), a figure CSV under `notebooks/analysis/figures_*`, or a pulled cluster artifact (cited relative to the local results tree `xcquinox-results/runs/`, e.g. `dfs_step7/dfs6311_grid3_v6g1_size/runs/run_-20260827T163330Z/...`). Values re-derived here were evaluated directly from the named artifacts and committed implementations. Coverage is stated per number; the campaign is live and several sets below are partial. Current as of 2026-09-01.

The functional family under study descends from Dick and Fernandez-Serra, Phys. Rev. B 104, L161109 (2021) ("DFS" below): 3x16 multilayer perceptrons returning exchange and correlation enhancement factors over an LDA baseline, trained through a differentiable self-consistent field. The meta-GGA rung uses the iso-orbital indicator of SCAN (Sun, Ruzsinszky and Perdew, Phys. Rev. Lett. 115, 036402 (2015)).

Figures are embedded at the point their subject is introduced, each with the reading it supports. All figure paths are relative to `notebooks/analysis/`, where this file lives and where the document is rendered.

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1. The campaign in one page

This section fixes the vocabulary the rest of the document uses. A reader who knows density functional theory but not this codebase needs it; a reader who knows the codebase can skip to Section 2.

1.1 Architecture, cell, spec

An **architecture** is a named entry of the registry `xcquinox/alec/config.py ARCHITECTURES` (31 entries, `config.py:505-684`). It fixes the network shape (hidden layers x width, with or without attention) and the **descriptor set** – the extra per-grid-point columns the two MLPs read beyond the semilocal variables (Section 3).

A **grid cell** is the swept unit: `GridCell` (`cluster/grid_config.py:60-72`) carries five fields, `(arch, loss, metric, subset_size, solver)`, and the grid is their Cartesian product in that fixed axis order, the position in the expansion being the SLURM array task index (`grid_config.py:1656-1673`). In every dfs6311 campaign configuration three of the five axes are singletons – `loss: L5_gradnorm_vxc_step7`, `metric: jsd`, `solver: full_3` – so a cell is in practice exactly one **(architecture, training-subset-size)** pair, and the grid arithmetic is stated per file (e.g. `hpcjobs/configs/dfs_step7.dfs6311_grid3_v6g1_size.yaml:105-110`, "4 x 1 x 1 x 11 x 1 = 44 cells").

A **spec** is the on-disk materialization of one cell: a pickled `TrainingSpec` (`config.py:961` onward) written as `spec_<index>.spec` with a SHA-256 recorded in the run's `manifest.json` (`cluster/materialize.py:123-268`), and a checkpoint directory `<run>/checkpoints/spec_<index>/`. What does **not** vary per spec is as important as what does: the random seed is a single run-level value (`HyperParams.seed = 42`, `grid_config.py:168`, threaded verbatim at `cluster/spec_builder.py:631`), as are the basis and grid level (`grid_config.py:211-212`). There is no replicate axis anywhere in the harness, so **every point on every curve below is a single training run**, and cell-to-cell scatter at fixed architecture carries seed variance that the campaign does not resolve. What *is* derived per spec is the SCF seed density: under `inputs.seed_xc: auto` the source is resolved from the architecture's rung (`spec_builder.py:571-577`), converged PBE for GGA architectures and converged SCAN for meta-GGA ones (Section 5).

1.2 The training pool and the subset-size axis

Training points are drawn from the DFS Letter's own training pool, transcribed verbatim from its supplementary material (`xcquinox/alec/dfs_pool.py:1-12`, "SI Sec. II 'Training Data' "): 21 atomization energies from G2/97

(10 linear closed-shell, 3 linear open-shell, 8 non-linear), 3 BH76 reaction barriers, 2 IP13 ionization potentials, and 2 atomic-density references (H, Li) that are always present rather than swept. The 26 selectable points are assembled by `training_points.build_dfs_pool_points` (`training_points.py:312-316`).

Subset size (ss) counts training points, not molecules and not reactions. The axis is `[1, 2, 3, 4, 5, 6, 7, 12, 15, 18, 26]` in all thirteen dfs6311 configurations (e.g. `dfs_step7.dfs6311_grid3_v6g1_size.yaml:129`), the largest equal to the whole pool. The composition at each size is fixed once, in the committed ledger `notebooks/checkpoints_step7/alpha_on/subset_index_log.json`, and reused by every generation – which is what makes the subset axis comparable across the campaign lineage (`CAMPAIGN_V6.md:695-700`). At `ss = 12`, for instance, the ledger's `jsd/12` entry holds 7 atomization energies, 3 BH76 barriers and 2 ionization potentials; at `ss = 26` it holds all 21, 3 and 2. Selection itself is a pre-process, not part of the harness: `subset_selection.select_subset` (`subset_selection.py:472-487`) enumerates every size-r combination and minimizes the Jensen-Shannon divergence between the subset's and the full pool's histograms over $(\rho^{1/3}, s, \alpha)$; the harness consumes the resulting ledger by point *name*, never by index (`spec_builder.py:38-43`). The number of molecules per spec is larger than ss and varies, being the union of every chosen point's reactants, products and atomic anchors (`spec_builder.py:502-511`).

1.3 The held-out pools and their slices

Two GMTKN55 subsets are held out (`notebooks/analysis/HOLDOUT_SET.md:317-324`):

- **BH76** – forward barrier heights (Goerigk, Hansen, Bauer, Ehrlich, Najibi and Grimme, Phys. Chem. Chem. Phys. 19, 32184 (2017)), 76 reaction entries.
- **W4-11** – zero-point-exclusive non-relativistic atomization energies (Karton, Daon and Martin, Chem. Phys. Lett. 510, 165 (2011)), 140 entries.

Their union is 216 reaction entries over 214 unique species, with 17 species shared (`full_benchmark_pools.py:516-527`). Two distinct multiplicities reduce that count and should not be conflated. Four BH76 entries are *listed twice under the same name* – `bh76_fch3fcomp_to_fch3fts`, `bh76_clch3clcomp_to_clch3clts`, `bh76_h_H2_to_RKT06` and `bh76_C5H8_to_RKT22` – so 216 entries carry **212 distinct names**. Separately, several barriers appear under permuted reactant order, so the 216 entries carry only **208 distinct physical identities** (both counts re-derived by enumerating `load_full_held_out_pools()` and `reaction_identity_key`).

Three slices matter, and the document keeps them apart:

1. **Validation.** A deterministic 20% partition – **34 names, 35 entries** (15 BH76, 20 W4-11) – keyed on the reaction's *physical identity* (sorted case-folded reactant and product tuples), not its name, so permuted-name twins land on the same side (`eval_holdout.split_held_out`, `eval_holdout.py:174-198`; `reaction_identity_key`, `:158-166`). It drives in-loop early stop and validation-best checkpoint selection and is withheld from every reported metric.
2. **Test ("strict").** The complement, **178 names covering 181 entries** (61 BH76, 120 W4-11), minus each cell's own *verbatim supervised* reactions. The exclusion is verbatim, not species-level: a held-out reaction that merely contains a trained molecule stays, because it is a generalization target (`eval_holdout.trained_reaction_exclusion`, `:237-292`; `_finalize_holdout_outputs`, `:1304-1330`). Bridging the two naming vocabularies – ASE Hill formulas in the training pool (`HO`, `CHN`, `H3N`) against GMTKN55 names in the benchmark pools (`oh`, `hcn`, `nh3`) – is an explicit identity layer keyed on `(composition, charge, spin)` (`species_matching.py:1-24`), without which a strict filter keeps trained molecules' reactions in the "held-out" set.
3. **In-sample.** The cell's own training subset, reported separately.

Because the exclusion is per cell, slices are nearly but not exactly uniform across subset sizes, and they can only shrink from the 61 / 120 / 181 the split itself defines: over the 27 completed v6 G1 cells the strict slices hold 61 BH76, 111–120 W4-11 and 172–181 combined reactions; over the 54 v4 GGA cells, 43–50, 97–113 and 145–163. Every "beats PBE" verdict below is therefore **within-cell** – the NN mean absolute error against the PBE mean absolute error on exactly the reactions the network was scored on – never a cross-campaign row-for-row comparison.

The density channel is per species rather than per reaction: each evaluated non-atomic species carries `density_rmse` (network against CCSD at the run's own basis and grid) and `density_rmse_pbe` (the model-free PBE twin). Atomic species are skipped by design, since the atomization anchors make lone-atom densities redundant, leaving **199 density species** of the 214 evaluated (counted from the `density_rmse` column of a completed spec's `per_molecule.json`; `HOLDOUT_SET.md:304-313`).

Two conventions used throughout are worth naming here rather than at first use. **WTMAD-2** is GMTKN55's weighted total mean absolute deviation, the benchmark's own cross-subset aggregate: each subset's mean absolute error is rescaled by the ratio of the average absolute reference value across all subsets to that subset's own, so that subsets whose reference energies are intrinsically large do not dominate a single number (Goerigk et al. 2017, cited via the repo record at `notebooks/analysis/README_density_figures.md`). It is reported in kcal/mol and is **not** the same quantity as the plain mean absolute error of Section 9. The **orientation lock** is a traceless-quadrupole bias of strength 3×10^{-5} (dimensionless) added to the core Hamiltonian of every reference and evaluation SCF; it removes the arbitrary spatial orientation a degenerate open-p-shell atom's converged density would otherwise pick at random, which is what makes the atomic references reproducible across runs and machines (HISTORY 2026-07-05).

1.4 The four evaluation channels

Every completed spec is evaluated four times. All four are dispatched from `_run_held_out_eval` in `cluster/_eval_one_spec.py`; the channel-to-checkpoint map is stated verbatim at `hpcjobs/reeval_c2_patch.py:177-186`.

channel	checkpoint	what selects it	SCF protocol
`eval_holdout` (final)	`model.eqx`	the last training step, written unconditionally (`train.py:915`)	the spec's own `full_3` solver: FULL mode, 3 cycles, `REASSEMBLE` features (every density-matrix-dependent descriptor is recomputed from the live density matrix each SCF cycle, rather than held at its entry value), decaying-linear mixer, `conv_tol` 10^{-6}; seed = converged parent density
`eval_holdout_best` (best-loss)	`model_best.eqx`	minimum trailing-mean **training** loss over one epoch (`train._BestModelTracker`, `train.py:537-543`)	as above
`eval_holdout_val_best` (validation-best)	`model_val_best.eqx`	minimum held-out **validation** metric, checked every 25 steps with patience 5 (`train._BestValidationTracker`, `train.py:566-575`)	as above
`eval_holdout_coldstart`	`model.eqx`	the same final weights under a different SCF	functional-free `minao` seed, 25 cycles (the Letter's step count), `conv_tol` 10^{-12} so the latched freeze never masks the trajectory (`eval_holdout.py:49-75`)

Two of these carry standing caveats. The **best-loss** channel minimizes a quantity that keeps falling as a network overfits, so on overfit-prone architectures it selects the *most* overfit snapshot; it was measured worse on held-out data than even the final step for the v3 `deep_combined` specs (99.98 against 70.62 kcal/mol at spec_0024) and is no longer plotted (HISTORY 2026-06-28; `make_ablation_arch_figure.py:7531`). The **cold-start** channel is a trajectory diagnostic, not a converged-evaluation replica: the manual solver has linear mixing only, so its 25 recorded cycles show whether a functional is walking toward a fixed point rather than certifying that it arrived (HISTORY 2026-08-14). It exists only from v5 onward (`grid_config.py:567-572`). Everything headline in this document is the **validation-best** channel, which is what the figure suite plots (`make_ablation_arch_figure.py:8706-8707`).

1.5 Generations and runs

The table below is the roster the rest of the document refers to by short name. Three of those names recur often enough to fix here: **v3** is the earliest sweep at DFS reference parity and supplies the oldest comparison curves; **G1** is the v6 anchored size ladder (`dfs6311_grid3_v6g1_size`), the only v6 group with trained cells on disk and therefore the source of every v6 held-out number below; and **`merged_v4_arms`** is not a campaign but a derived directory that merges the completed v4 and v4gga arms onto one comparison slice, which is what the "merged" figure sets are built from (Sections 9.4 and 10.3).

One recurring failure mode is also named once here. The **open NaN-gradient defect** is an unresolved condition in which a training cell's loss gradient becomes non-finite mid-run and the cell terminates without writing a usable checkpoint; it is what left the G1 `medium` ss=26 cell absent from the tables below, and it is open rather than diagnosed (HISTORY 2026-08-31).

generation	architectures	cells	distinguishing feature
v3	8 (GGA + first meta-GGA forms)	88	first sweep at DFS reference parity (6-311++G(3df,2pd), grid 3); closes as the pre-V_{xc}-correction record
v4	3 meta-GGA	33	meta-GGA arm of the corrected-V_{xc} re-sweep
v4gga	6 GGA	66	GGA arm of the same re-sweep, same methodology, separate output root
v4mgga2	2 meta-GGA stacking forms	22	the two registry completions added 2026-08-10
v5	3 meta-GGA	33	SCAN-seeded retrain of the v4 meta-GGA arm; a controlled seed A/B, the v4 rows remaining as the PBE-seeded control
v5mgga2	2 meta-GGA	22	the same seed A/B on the stacking arm
v6 G1 (`v6g1_size`)	shallow, shallow__attn, medium, medium__attn	44	anchored size ladder: does capacity limit anything at fixed inputs
v6 G2a (`v6g2a_families_core`)	deep__3x16, deep__attn__3x16, deep__cusp__3x16	33	anchored GGA core trio
v6 G2b (`v6g2b_families_rung35`)	the three rung-3.5 GGA forms	33	anchored rung-3.5 trio
v6 G2 meta-GGA (`v6g2_families_mgga`)	five meta-GGA family forms	55	SCAN-parented, SCAN-seeded parity forms
v6 G3 (`v6g3_dm`)	deep__dm__3x16, deep__combined__3x16, deep__combined__attn__3x16	33	the density-matrix indicators in their repaired two-column form
v6 G4 (`v6g4_ablations`)	deep, deep__attn	22	depth 4 / width 32 against the production depth 3 / width 16

v6 totals 220 cells over 20 of the 31 registry architectures (`CAMPAIGN_V6.md:934-940`). The file `dfs_step7.dfs6311_grid3_v6.yaml`, covering all 31 architectures at 341 cells, is the statement of the method and is not submitted (`CAMPAIGN_V6.md:934-940`).

1.6 From artifacts to figures

Each held-out channel writes three data files plus a provenance stamp (`eval_holdout._finalize_holdout_outputs`, `eval_holdout.py:1379-1387`):

- `test_set.csv` – the per-spec mean-absolute-error summary, one row per pool plus one combined, with `n_reactions`, `n_dropped_overlap` and `n_dropped_nan` (`eval_holdout.write_test_set_csv`, `:875-964`). **The tables of Sections 9.2 and 9.4 are derived from these files.**
- `per_reaction.json` – one record per surviving reaction: reference energy, network and PBE reaction energies and absolute errors, and the annotated `in_sample_overlap` list.
- `per_molecule.json` – one record per evaluated species: energies, the six density-error columns (`density_rmse`, `density_l1` and `density_eps_l1`, each with its `_pbe` twin), SCF convergence and cycle count.
- `eval_metadata.json` – channel name, checkpoint filename, cold-start flag, the serialized solver configuration, and the species slice (`_eval_one_spec.py:471-489`).

The **figures** are built by `notebooks/analysis/make_ablation_arch_figure.py` and read the two JSON files, not the CSV: `collect_holdout_reaction_rows` (`:274-386`) reconstructs each spec's full test slice from the per-species energies of `per_molecule.json` and re-applies the verbatim and validation exclusions itself, falling back to `per_reaction.json` only for pulls that predate the energy columns. `collect_holdout_density_rows` (`:4010`) reads the density columns. The enhancement-factor figures come from two further modules, `pretrain_fx_fc.py` and `trained_fx_fc.py`, which load the stored network weights through the production certified-model builder and evaluate them on a fixed (r_s, s) grid. Each of those two modules writes **one** long-form CSV per figure directory – `pretrain_fx_fc_curves.csv` and `trained_fx_fc_curves.csv`, one row per plotted grid point across all architectures and cells – so every plotted enhancement-factor number is recoverable without re-running the code. The `ablation_*` suite is different: only some of its figures have a same-stem CSV, and the three `ablation_*` figures embedded below (Section 9.2) have none, so their captions quote the per-spec artifacts they are built from instead.

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2. The common frame: networks, bounded maps, parent conventions

All three generations share the network form (`xcquinox/alec/networks.py`). Per grid point the exchange network reads the reduced gradient $s = |\nabla\rho|/(2k_F\rho)$ of one spin channel posed on its doubled density, and the correlation network reads the total density with the spin polarization ζ; descriptor-carrying architectures append extra columns (cusp, rung-3.5 occupancies, the meta-GGA indicator; Section 3). The raw MLP output is gated by a UEG-recovery prefactor ($\tanh^2 s$ at the GGA level; the DFS Eq. 12 form $(x_2 + \tanh^2 x_3)$ at the meta-GGA level) and passed through a bounded squash with a static limit (`networks._AlecLOB`, `networks.py:65`):

$$L(x) = \Lambda\,\sigma(x - \ln(\Lambda - 1)) - 1,$$

with σ the logistic function; $\Lambda = 1.804 = 1 + \kappa_{\mathrm{PBE}}$ for the GGA exchange ceiling; $\Lambda = 1.174$, the DFS exchange ceiling, for a meta-GGA exchange network – hardcoded in `networks.create_network_pair` (line 751), bypassing the registry's 1.804, and touched exactly by the parent, since SCAN's F_x reaches exactly 1.174 at $s = 0$, $\alpha = 0$, where the pre-image clamps; and $\Lambda = 2.0$ for the correlation squash, whose purpose is non-negativity of F_c (the DFS Eq. 13 transform), not a Lieb-Oxford bound (`networks._AlecLOB` docstring). Writing $T(g)$ for the gated network term on the descriptor row g:

- **Unanchored** (v3/v4/v5): $F = 1 + L(T(g))$. With the zero-initialized final layer the networks start at $T(g) = 0$, i.e. $F \equiv 1$ (the LDA limit), and pretraining moves the curve toward the parent.
- **Anchored** (v6): $F = 1 + L(z_{\mathrm{parent}} + T(g))$ with $z_{\mathrm{parent}} = L^{-1}(F_{\mathrm{parent}} - 1)$, so $T(g) = 0$ returns the parent exactly (Section 6.1).

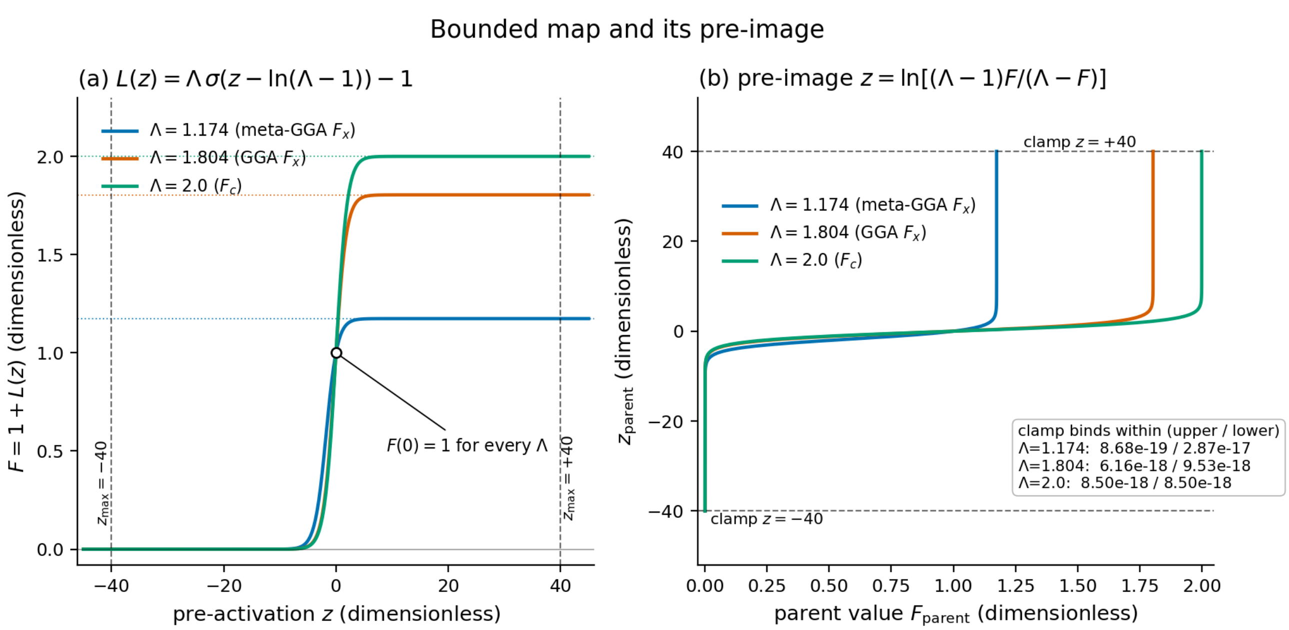

Figure 1: Bounded map and its pre-image. Left: $F = 1 + L(z)$ for the three limits in use, with the pre-image clamp at $z = \pm 40$ marked. Right: the inverse $z = \ln[(\Lambda - 1)F/(\Lambda - F)]$, annotated with the distance from each bound inside which the clamp binds.

Both panels are drawn by calling `networks._AlecLOB` and `parents.lob_preimage` themselves (`report_equation_figures.make_bounded_map`), so the figure cannot drift from the code. The left panel is the constraint: whatever the MLP emits, F stays inside $(0, \Lambda)$, and every curve passes through $F = 1$ at $z = 0$ – the property that makes a zero-initialized final layer the LDA limit for an unanchored network and the parent for an anchored one. The right panel is the price: the map is exponentially flat near both bounds, so a parent within $\Lambda(\Lambda-1)e^{-40}$ of the ceiling or $\Lambda e^{-40}/(\Lambda-1)$ of the floor cannot be represented by a finite pre-image and is clamped instead. The figure's own bind thresholds, computed from that closed form and then checked against `lob_preimage` by requiring the clamp to return exactly ± 40, read (from `bounded_map.csv`, panel `c_bind`) 8.678×10^{-19} / 2.866×10^{-17} at $\Lambda = 1.174$, 6.162×10^{-18} / 9.532×10^{-18} at 1.804, and 8.497×10^{-18} at both bounds of $\Lambda = 2.0$. **Conclusion:** the clamp is inert at any physical parent value; the flatness it reflects is not, and Section 8.2 is about exactly that flatness.

The parent is fixed per rung: PBE (Perdew, Burke and Ernzerhof, Phys. Rev. Lett. 77, 3865 (1996)) for GGA-rung architectures, SCAN for meta-GGA ones (`parents.parent_for_arch`; `cluster/fidelity.resolve_parent` reads the same predicate). Exchange is posed per spin channel on the doubled density, following the exact spin scaling $E_x[\rho_\alpha, \rho_\beta] = (E_x[2\rho_\alpha] + E_x[2\rho_\beta])/2$ (Oliver and Perdew, Phys. Rev. A 20, 397 (1979)); correlation is posed on the total density with ζ, its target ratio formed against the polarized PW92 baseline (Perdew and Wang, Phys. Rev. B 45, 13244 (1992)) the model itself multiplies (`parents.py` module docstring).

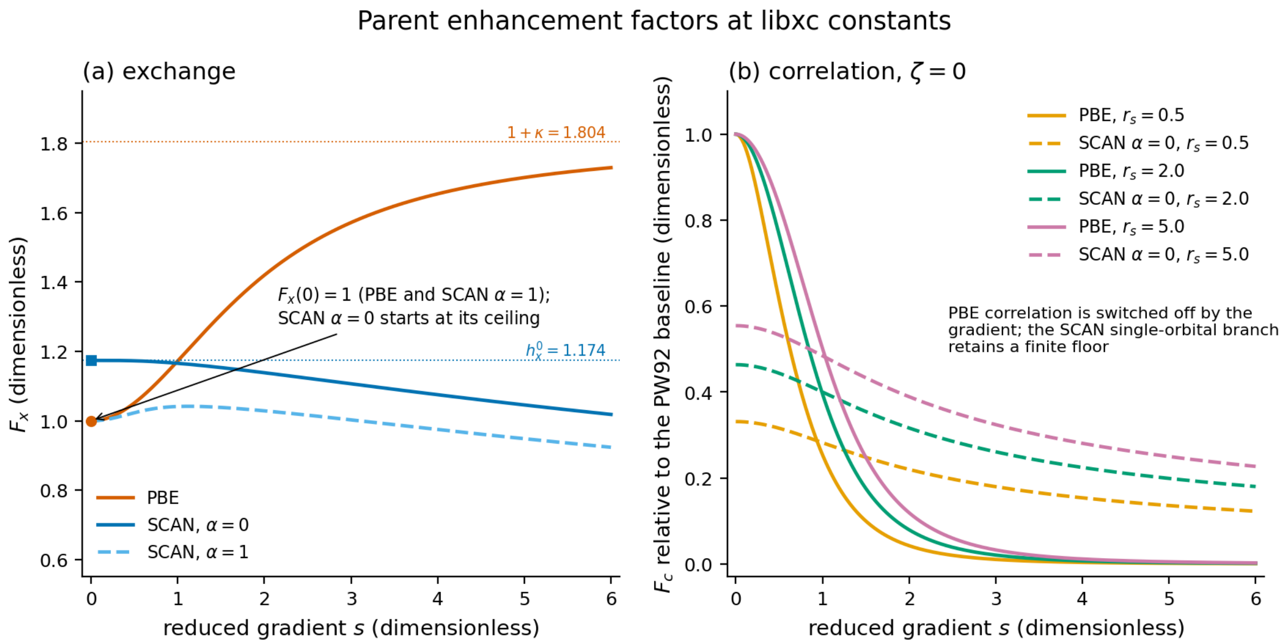

Figure 2: Parent enhancement factors at libxc constants. Left: exchange, PBE against SCAN at $\alpha = 0$ and $\alpha = 1$, with the two ceilings marked. Right: correlation relative to the PW92 baseline at three densities, $\zeta = 0$.

These are the curves every "distance from the parent" statement in this document is measured against, evaluated through `parents.pbe_fx` / `pbe_fc` / `scan_fx` / `scan_fc` at libxc's own constants. Read the left panel for the two exchange conventions: PBE rises monotonically from $F_x(0) = 1$ toward its ceiling (`parent_enhancement.csv` gives $F_x = 1.72976$ at $s = 6$, still below 1.804), SCAN at $\alpha = 1$ likewise starts at 1, while SCAN at $\alpha = 0$ – the single-orbital branch – *starts at its ceiling*, exactly 1.174. Read the right panel for the correlation asymmetry that Section 8.2 turns on: PBE correlation is switched off by the gradient, falling from $F_c \approx 1$ at $s = 0$ to 1.532×10^{-3} at $s = 6$ for $r_s = 2$ (and to 7.56×10^{-4} at $r_s = 0.5$), whereas SCAN's single-orbital branch retains a finite floor, 0.18019 at the same point. **Conclusion:** the two parents differ qualitatively in the large-gradient correlation limit, and the anchored parameterization's sensitivity is proportional to the parent's own value there – so the same anchor costs more against PBE than against SCAN.

The correlation baseline itself carries a coordinate singularity worth stating, because it is clipped rather than removed:

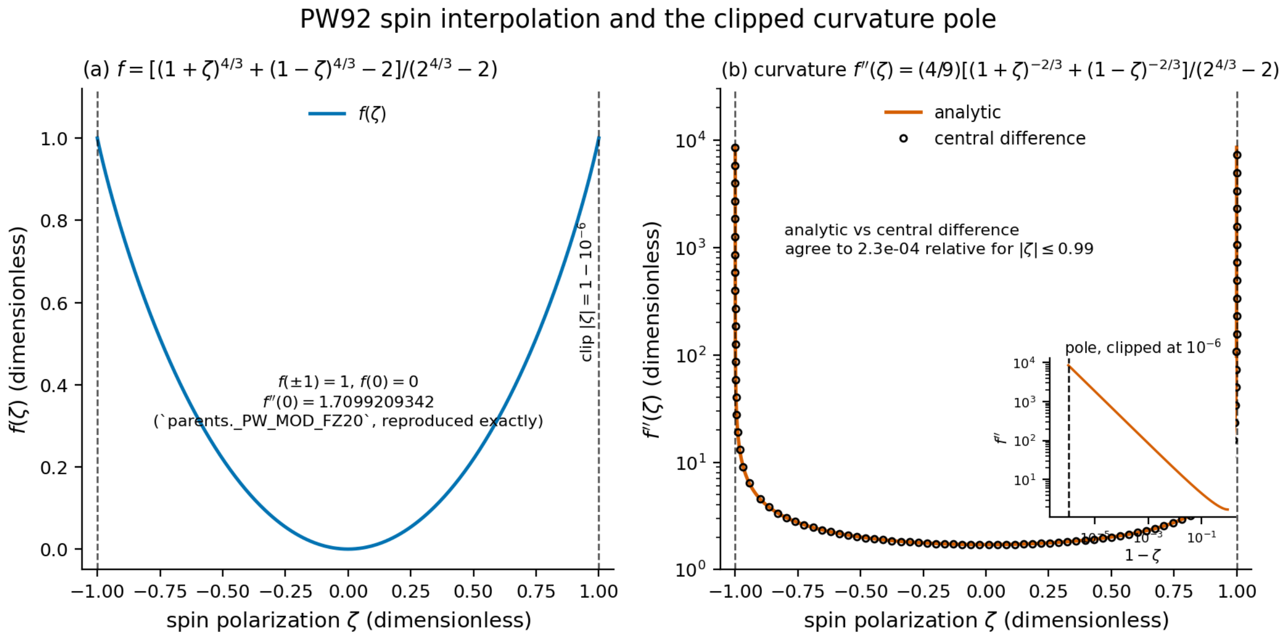

Figure 3: PW92 spin interpolation $f(\zeta)$ and its curvature. Left: the interpolation, with the production clip at $|\zeta| = 1 - 10^{-6}$ marked. Right: $f''(\zeta)$, analytic against a central difference, on a log scale, with the pole resolved in the inset.

$f(\zeta) = [(1+\zeta)^{4/3} + (1-\zeta)^{4/3} - 2]/(2^{4/3} - 2)$ is the PW92 spin interpolation (Perdew and Wang, eq. 9). It is smooth, but its second derivative diverges as $|\zeta| \to 1$: `zeta_pole.csv` records $f''(0) = 1.7099209342$ – reproducing `parents._PW_MOD_FZ20` exactly, which the figure asserts as an oracle – rising to 8.550×10^3 at the clip $|\zeta| = 1 - 10^{-6}$. A fully polarized spin channel therefore has an unbounded correlation second derivative, and taking one through a full SCF returns non-finite values; the production path clips ζ at $1 - 10^{-6}$ (`oneshot._ZETA_BOUNDARY_EPS`). The written f and its analytic f'' are not independent assertions: the figure reconstructs `parents._pw92_mod_eps` from them through the repository's own $G(r_s)$ parametrizations and raises if the agreement exceeds 10^{-15} (`report_equation_figures._verify_f_spin`), and the drawn central difference tracks the analytic form to 2.3×10^{-4} relative over $|\zeta| \leq 0.99$ – the figure's own stated agreement, dominated by the stencil's truncation error where the curvature is steepest; over the flat interior $|\zeta| \leq 0.5$ the same comparison closes to 2.6×10^{-7}. **Conclusion:** the clip is a numerical necessity of the baseline, not of the network, and it is the reason an anchored correlation network must be polarization-aware (Section 6.1).

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3. The descriptors

Beyond (ρ, σ) for exchange and (ρ, ζ, σ) for correlation, an architecture may append **descriptor** columns to both MLP inputs. Each descriptor is a registered class in `xcquinox/alec/descriptors.py` implementing a `compute` method that returns an $(N_{\text{grid}}, n_{\text{features}})$ block, and `assemble_descriptor_features` (`descriptors.py:475-494`) concatenates the blocks left to right in the architecture's declaration order. Column order and width are identical whether the block is evaluated on the physical (total) density, which the correlation term consumes, or on the symmetric doubled density $\text{diag}(P_\sigma, P_\sigma)$ of one spin channel, which the exact exchange spin scaling requires.

The five registered descriptors follow. Ordering within an architecture is not canonical – the `deep_combined` family declares `("dm_statistics", "cusp")` while every rung-3.5 and meta-GGA family declares cusp first – so the input-column layout differs between families and cannot be assumed alphabetical.

3.1 `cusp` – nuclear cusp proximity

`CuspDescriptor` (`descriptors.py:226`, `compute` at `:257`) supplies two bounded features per grid point:

$$c_0(\mathbf{r}) = \exp(-2Z_{\text{near}}\, r_{\min}) \in [0,1], \qquad c_1(\mathbf{r}) = \tanh\left[\tfrac{1}{5}\,\ln\left(\sum_A \frac{Z_A}{r_A} + 10^{-12}\right)\right] \in (-1,1),$$

with Z_{near} the charge of the nearest nucleus, $r_{\min}$ the distance to it, the sum in c_1 over all nuclei, and the 10^{-12} guarding the logarithm at a vanishing weighted sum (`features.py:329`). Under `log_transform=False` the logarithm is dropped and $c_1 = \tanh(\frac{1}{5}\sum_A Z_A/r_A)$; the two are semantically distinct checkpoint families, and the trained checkpoint's class record now states `descriptor_log_transform` and refuses a load whose record and skeleton disagree on it (`checkpoint_class.require_matching_log_transform`); a record written before the field existed – every cluster record at the time of writing – is accepted exactly as before, the check firing only when both sides state the flag. Measured consequence of a silent cross-flag load, the case the record now refuses: the cusp c_1 column moves by 0.51 on its bounded $(-1,1)$ range at 0.3–4 bohr from an oxygen nucleus.

Which form a run uses is not a descriptor-local setting, and the routing matters later. The architecture-level `descriptor_log_transform` reaches **two** distinct places. One is the MLP coordinate transform of Section 3.7, which the DFS coordinate set bypasses entirely. The other is this logarithm: `ArchitectureConfig.materialize_descriptors` (`config.py:351-368`) copies the arch-level flag into the cusp descriptor's own `log_transform` kwarg whenever the feature spec does not set it explicitly. That second path is live under *every* coordinate set, DFS included, so a flag that is inert for a descriptor-free architecture is not inert for a cusp-carrying one. Over a whole converged PBE H2O density at def2-svp (30,632 grid points) the two settings place c_1 at -0.026 to 0.981 and at 0.174 to 1.000 respectively, differing by up to 0.534 point-wise, while c_0 is bitwise identical. Section 9.1 turns on exactly this distinction.

The rationale is the electron-nucleus cusp. Kato (*Commun. Pure Appl. Math.* 10, 151 (1957)) fixes the wavefunction condition $(\partial\langle\psi\rangle/\partial r)|_{r=0} = -Z\psi(0)$; the corresponding spherically averaged density relation $(\partial\langle\rho\rangle/\partial r)|_{r=0} = -2Z\rho(0)$ is due to Steiner (*J. Chem. Phys.* 39, 2365 (1963)). Column c_0 approximates the resulting Slater envelope $\exp(-2Zr)$ rather than enforcing the condition; the docstring states this explicitly, and the feature should be described as a proximity heuristic motivated by the cusp, not as a cusp constraint. Column c_1 is a log-compressed nuclear-attraction weight in the DFS convention, the 1/5 scaling chosen on a dynamic-range argument (`xcquinox.features.compute_cusp_descriptor`). Both are pure geometry: they do not depend on the density matrix, and they are exact under the doubled-spin substitution, which is why the cusp architectures' pretraining offsets of Section 4.3 are *not* spin-scaling defects.

Carried by: `deep_cusp`, `deep_cusp_attn`, `deep_cusp_3x16` (cusp alone); the four `deep_combined*` forms; both `deep_rung35*` forms and `deep_rung35ms_3x16`; and **three of the five** meta-GGA forms (`deep_cusp_mgga_3x16`, `deep_rung35_mgga_3x16`, `deep_rung35ms_mgga_3x16`; `deep_mgga_3x16` and `deep_mgga_attn_3x16` carry the indicator alone).

3.2 `dm_statistics` – global density-matrix indicators

`DMStatisticsDescriptor` (`descriptors.py:262`, `compute` at `:315`) supplies two features (`features.compute_dm_features_array`, `features.py:109-181`):

$$\text{idem} = \frac{\|PSP - P\|_F^2}{N_{\text{pair}} + 10^{-12}}, \qquad \text{offdiag} = \frac{\|D - \text{diag}(D)\|_F}{\operatorname{Tr} D + 10^{-12}}.$$

Two details of the first are load-bearing. The operand is the **pair** density matrix $P = D/2$, normalized by the **electron-pair** count $N_{\text{pair}} = \text{Tr}(PS) = N_e/2$, which is what makes $PSP = P$ the idempotency condition at all; on an open-shell UKS row the feature is instead the per-spin mean $\frac{1}{2}[\text{idem}(P_\alpha) + \text{idem}(P_\beta)]$ (`features.py:113-127`). And the norm is **squared**: $\|X\|_F$ is not differentiable at $X = 0$, which is identically where every converged density sits, and autodiff there returned -2.08×10^{-3} against a finite difference of $+4.5 \times 10^{-9}$; the squared form is a polynomial with the same zero set and restores agreement to 6.5×10^{-15}.

Only the first feature vanishes on a single determinant. `idempotency_error` is zero to round-off at any exact Hartree-Fock or Kohn-Sham reference and grows with departure from a single Slater determinant, but `off_diag_norm` does **not** – it measures AO-basis off-diagonality, which a single determinant has in abundance. Measured on a converged closed-shell PBE H2O reference (def2-svp), whose $\|PSP - P\|_F$ is 1.15×10^{-15}: `idempotency_error` reads 2.65×10^{-31} while `off_diag_norm` reads **0.2714**. The second feature is a bonding-structure indicator, not a correlation indicator that vanishes at the mean-field limit.

Two properties must be stated with the definition. First, these are **global, per-molecule scalars**, `jnp.tile`d identically to every grid point and fed into a per-point (semilocal) enhancement factor – so the exchange-correlation energy density at a point in fragment A shifts when a distant fragment B is added. The size-consistency and locality caveat is recorded in the class docstring (`:288-296`) and remains open; the rung-3.5 descriptors below are the leak-free members of the same family. Second, a third feature, `dm_entropy`, was **removed on 2026-08-06** (width 3 to 2): it had no usable gradient at any converged density, because the physical-bounds clip put every natural occupation on a boundary and, without the clip, the eigenvector derivatives of `eigh` are ill-defined on the degenerate occupation spectrum of an idempotent density matrix. No spectral invariant can replace it – for a single determinant the eigenvalues of DS are exactly $\{2, \ldots, 2, 0, \ldots, 0\}$, so any function of the spectrum alone is constant on the idempotent manifold. Removing it took the energy/potential finite-difference residual of the dm_statistics architectures from 1.04×10^{-2} to 2.1×10^{-10} under the committed test's own ordering, the dead gradient having dominated it (`descriptors.py:275-286`; `features.py:130-155`; `notebooks/analysis/DM_-DESCRIPTOR_SPEC.md`).

Carried by: `deep_dm`, `deep_dm_attn`, `deep_dm_3x16` (alone) and the four `deep_combined*` forms (with cusp). No meta-GGA architecture carries it.

3.3 `rung35` – localized density-matrix occupancy

`DMRung35Descriptor` (`descriptors.py:320`, `compute` at `:366`; kernel `xcquinox/alec/rung35.py:96`) is the bounded local occupancy of Janesko's unified rung-3.5 / DFT+U formalism (arXiv:2206.07118, Eqs. 12-13; M11plus, Verma et al., *J. Chem. Theory Comput.* 15, 4804 (2019)):

$$n_\sigma(\mathbf{r}_m) = \sum_i \left|\langle \psi_{i\sigma} \,|\, \phi^G_{\mathbf{r}_m} \rangle\right|^2 = A(\mathbf{r}_m)^\mathsf{T} P^\sigma A(\mathbf{r}_m) \in [0, 1],$$

with an L^2-normalized Gaussian projector centred at the grid point, $\phi^G_{\mathbf{r}_m}(\mathbf{r}) = (2\alpha/\pi)^{3/4} \exp(-\alpha|\mathbf{r} - \mathbf{r}_m|^2)$, and the projected-AO overlap vector $A_\mu(\mathbf{r}_m) = \langle \chi_\mu | \phi^G_{\mathbf{r}_m} \rangle$. The two features are the α- and β-spin occupancies, and they feed **both** networks: the M11plus rung-3.5 ingredient is a correlation ingredient, so the C-net is a first-class consumer, and the X-net receives it equally through the shared extras mechanism.

The construction is what makes the descriptor tractable. A_μ depends only on the basis, the grid and the *fixed* width α – not on the density matrix – so it is a precomputed constant (a plain PySCF overlap integral) that is never differentiated, and the occupancy is a single einsum, linear and differentiable in the *live* density matrix. The default width is `DEFAULT_RUNG35_ALPHA = 0.2` a_0^{-2} (`rung35.py:39`), which the repository states as grounded at the M11plus kernel scale $d^2 = 5\,a_0^2$; that attribution is carried by the repository itself and is still to be confirmed against the library copy of the paper (`CAMPAIGN_V6.md:229-230`), so it is reported here as the repo record rather than as a read of Verma et al. Boundedness in $[0, 1]$ follows from Bessel's inequality – P^σ is positive semidefinite, and orthonormal occupied orbitals against a normalized projector give the upper bound – so the feature is NaN-safe by construction. Unlike `dm_statistics` this is a genuine per-grid-point contraction of the non-local one-particle density matrix, hence size-intensive and leak-free, and it is self-consistent: a functional of the live density matrix, recomputed each SCF cycle under the `REASSEMBLE` policy, never a static reference. It carries no τ and is therefore not a meta-GGA – its own rung, between meta-GGA and hybrid.

Carried by: `deep_rung35_3x16` and `deep_rung35_attn_3x16` (with cusp), `deep_rung35only_3x16` (alone), and `deep_rung35_mgga_3x16` (with cusp and the indicator).

3.4 `rung35_multishell` – the radial generalization

`DMRung35MultishellDescriptor` (`descriptors.py:371`, `compute` at `:428`; kernel `rung35.py:156`) evaluates the same occupancy at several projector widths,

$$n_\sigma(\mathbf{r}; w) = A_w(\mathbf{r})^\mathsf{T} P^\sigma A_w(\mathbf{r}) \in [0, 1],$$

with DEFAULT_RUNG35_MULTISHELL_ALPHAS = (0.05, 0.2, 0.8) a_0^{-2} (rung35.py:130) – the M11plus scale bracketed by a factor of about four either side, so the set gives a coarse radial profile of the one-particle density matrix around each point. The feature count is $2\times$ the number of widths, six by default, ordered **alpha-major then spin**; setting a single width reproduces rung35 bitwise, and the constructor enforces the count relation (:413-418).

This is the radial part of the localized density-matrix projection used by NeuralXC (Dick and Fernandez-Serra, *Nat. Commun.* 11, 3509 (2020)) and carried in the DFS reference implementation, which projects the density matrix onto a localized basis and contracts the coefficients into rotationally invariant per-shell norms. The stated limitation belongs with the definition: fakemol_for_charges builds s-type projectors only, so only the $l=0$ channels exist, and with one m per shell the invariant $\sqrt{\sum_m c_{nlm}^2}$ collapses to the occupancy itself. Angular channels require solid-harmonic fakemols and are not implemented, so this should not be described as "the DFS descriptor" (descriptors.py:383-389).

Carried by: deep_rung35ms_3x16 (with cusp) and deep_rung35ms_mgga_3x16 (with cusp and the indicator).

3.5 metagga – the SCAN iso-orbital indicator

MetaGGAAlphaDescriptor (descriptors.py:433, compute at :471) supplies one feature, the iso-orbital indicator introduced by SCAN (Sun, Ruzsinszky and Perdew, Phys. Rev. Lett. 115, 036402 (2015), Eq. 2) and reused by DFS (Eq. 6):

$$\alpha = \frac{\tau-\tau_W}{\tau_{\text{unif}}}, \qquad \tau(\mathbf{r}) = \tfrac{1}{2}\sum_{\mu\nu} P_{\mu\nu}\,\nabla\chi_\mu\cdot\nabla\chi_\nu, \qquad \tau_W = \frac{|\nabla\rho|^2}{8\rho}, \qquad \tau_{\text{unif}} = \tfrac{3}{10}(3\pi^2)^{2/3}\rho^{5/3},$$

with $\alpha=1$ the uniform gas and $\alpha=0$ a single orbital. The kinetic-energy density is a *linear* contraction of the live density matrix against AO gradients already on the grid (metagga.compute_tau_from_dm, metagga.py:115), so – exactly like the rung-3.5 occupancy – the descriptor is self-consistent, differentiable through the SCF, and needs no new integrals, no Laplacian and no deriv=2.

The stored column is not α raw but

$$\min\bigl(p(\alpha_{\text{raw}}),\,100\bigr), \qquad p(x) = \tfrac{1}{2}\left(x+\sqrt{x^2+w^2}\right), \quad w = 10^{-5},$$

with p the smooth positive part (metagga.smooth_positive_part, :142; _ALPHA_SMOOTHING_WIDTH = 1e-5 at :104) and the ceiling _ALPHA_MAX = 100 at :61. The lower bound $\alpha\geq 0$ is the von Weizsacker inequality, exact on every positive semidefinite density matrix, so a negative raw value is rounding; the smoothing exists because a hard clip $\max(\alpha_{\text{raw}},0)$ has a one-sided derivative exactly on the manifold every one-electron spin channel occupies. Both regularizations are derived in Section 6.2 and Section 6.5 and are not repeated here.

Both panels call metagga.compute_alpha directly. Read the left panel for the two physical anchors: at $\tau=\tau_W$ the indicator goes to its smoothed floor rather than to zero (alpha_indicator.csv gives $5\times10^{-6}=w/2$ at raw indicator 0), and at $\tau=\tau_W+\tau_{\text{unif}}$ it reads 1.000000000025 – exactly $1+w^2/4$, the smoothing's own second-order offset at the uniform gas, which is the scale at which the regularization is visible at all. Read the right panel for the ceiling: over the four decades of raw indicator the panel draws (0.1 to 1000), the stored column tracks the raw value for the first three – to $\alpha=98.5$ – and then saturates, 151 of the panel's 601 points sitting at the ceiling (alpha_indicator.csv, panel b_ceiling). **Conclusion:** the column is the raw indicator everywhere the physics is resolved and departs from it only in two regimes – one below the numerical noise floor of the $\tau-\tau_W$ cancellation, one on the low-density tail where the indicator diverges. Section 6.5 prices both.

The left panel shows what the smoothing does: p equals $\max(x,0)$ to within $w^2/4|x|$ for $|x|\gg w$, sits at $w/2$ with slope 1/2 at the origin, is strictly positive everywhere, and satisfies $p(x)-p(-x)=x$ to one unit in the last place, so a central difference across zero reproduces its derivative. The right panel is what makes the anchored construction of Section 6 work at all: the stored column can be inverted, $x=p-w^2/(4p)$ (metagga.invert_smooth_positive_part, :155), and the round trip closes to 2.778×10^{-19} absolute over the plotted grid (8.470×10^{-15} relative), from smooth_positive_part.csv. **Conclusion:** below the ceiling the smoothing is invertible to

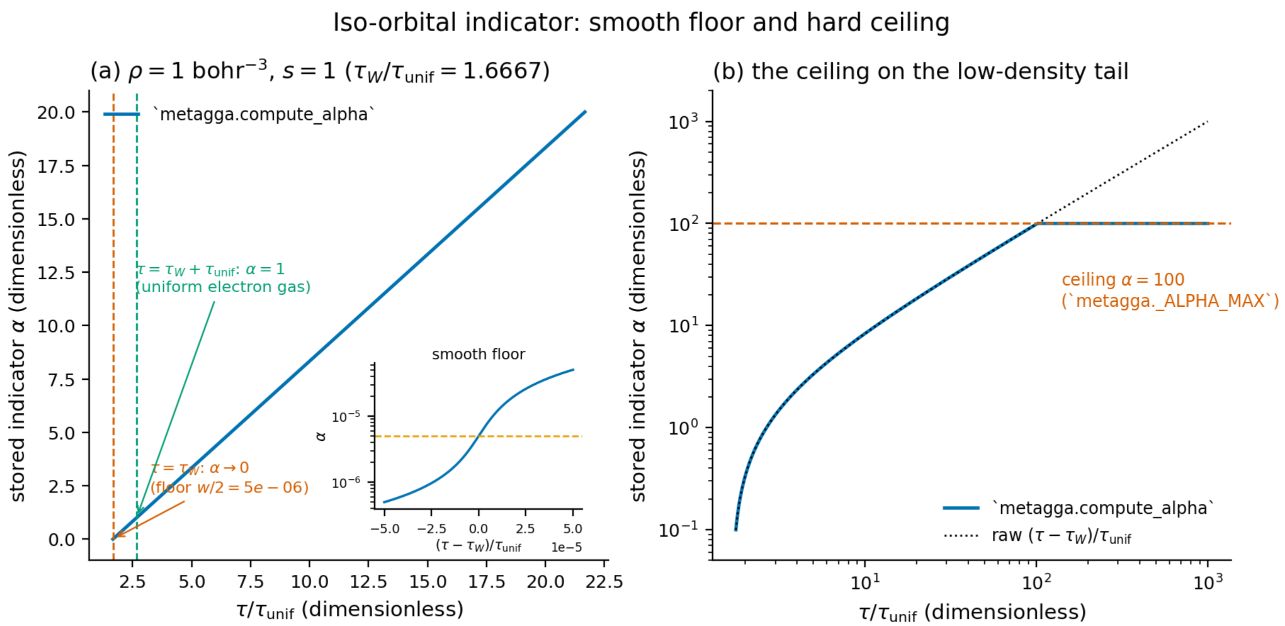

Figure 4: Iso-orbital indicator: smooth floor and hard ceiling. Left: the stored α against $\tau/\tau_{\mathrm{unif}}$ at $\rho = 1$ and $s = 1$, with the single-orbital and uniform-gas points marked and the smooth floor resolved in the inset. Right: the same on log axes, showing the ceiling truncating the low-density tail.

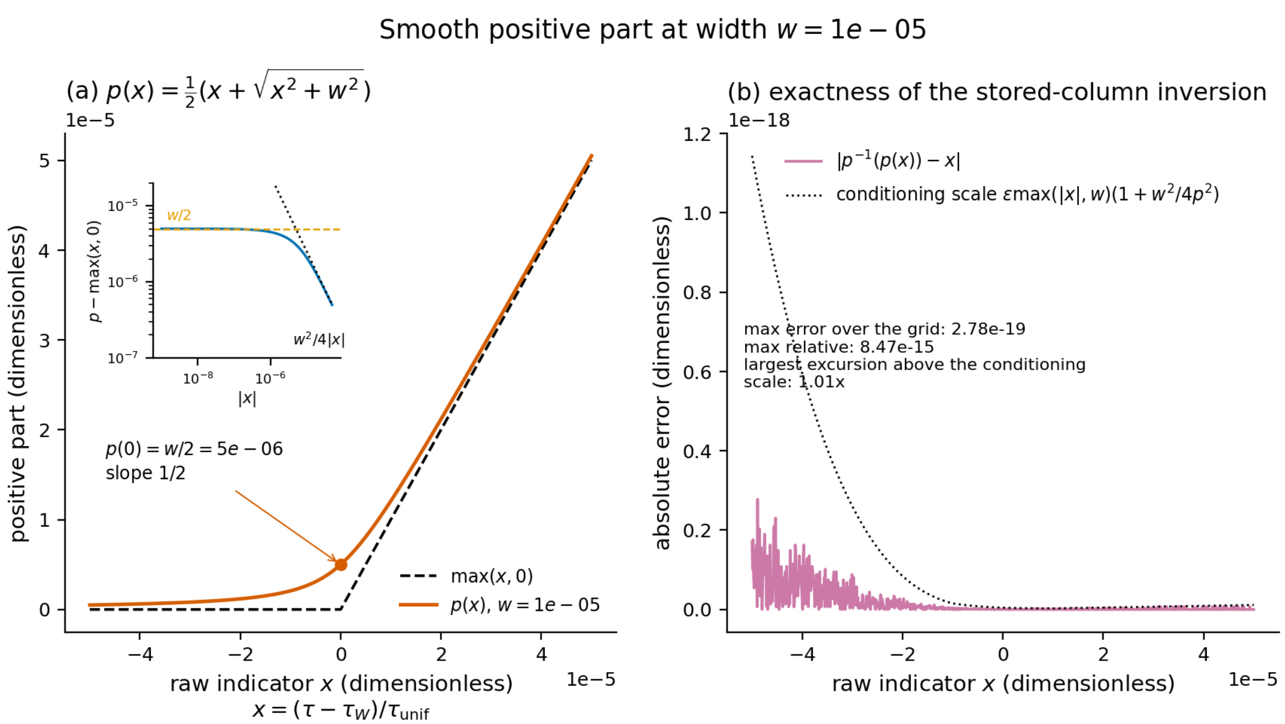

Figure 5: Smooth positive part at $w = 10^{-5}$. Left: $p(x)$ against the hard clip, with the excess over the clip and its $w^2/4|x|$ asymptote in the inset. Right: the exactness of the stored-column inversion, against a first-order conditioning scale.

round-off, so an anchored network recovers the exact indicator its parent needs; at the ceiling it is not, and that single fact is the whole content of Section 6.5.

One documented deviation belongs here. DFS feeds its network the log-transformed coordinate $x_3 = \ln((\alpha+1)/2)$ (its Eq. 10); in the legacy coordinate family the *raw* clamped column is the MLP input and x_3 enters only through the UEG-recovery gate. Under the DFS coordinate family (Section 3.7) the network input is x_3 itself. The deviation is recorded, not silently carried (`descriptors.py:450-452`).

Carried by: all five meta-GGA architectures.

3.6 The architecture-to-descriptor map

The registry holds 31 architectures. Grouped by descriptor set, with `config.py` line numbers (re-read from the current file):

descriptor set	count	architectures (line)
none	12	`shallow` (506), `shallow_attn` (507), `medium` (508), `medium_attn` (509), `deep` (514), `deep_attn` (518), `deep_notransform` (560), `deep_notransform_attn` (564), `deep_3x16` (573), `deep_attn_3x16` (577), `deep_notransform_3x16` (675), `deep_notransform_attn_3x16` (679)
`("cusp",)`	3	`deep_cusp` (523), `deep_cusp_attn` (528), `deep_cusp_3x16` (582)
`("dm_statistics",)`	3	`deep_dm` (534), `deep_dm_attn` (539), `deep_dm_3x16` (587)
`("dm_statistics", "cusp")`	4	`deep_combined` (545), `deep_combined_attn` (550), `deep_combined_3x16` (592), `deep_combined_attn_3x16` (597)
`("cusp", "rung35")`	2	`deep_rung35_3x16` (609), `deep_rung35_attn_3x16` (614)
`("cusp", "rung35_multishell")`	1	`deep_rung35ms_3x16` (624)
`("rung35",)`	1	`deep_rung35only_3x16` (628)
meta-GGA-bearing	5	`deep_mgga_3x16` (641) `("metagga",)`; `deep_mgga_attn_3x16` (646) `("metagga",)`; `deep_rung35_mgga_3x16` (652) `("cusp", "rung35", "metagga")`; `deep_cusp_mgga_3x16` (664) `("cusp", "metagga")`; `deep_rung35ms_mgga_3x16` (669) `("cusp", "rung35_multishell", "metagga")`

The twelve descriptor-free entries span the whole shape axis, which is what makes the size ladder of Section 9.1 a controlled comparison: `shallow` and `shallow_attn` are 2x8, `medium`, `medium_attn`, `deep_3x16` and `deep_attn_3x16` are 3x16, and `deep` and `deep_attn` are 4x32, with attention heads 2 for `shallow_attn` and 4 for every other attention form. One registry inconsistency is worth recording because it looks material and is not: `deep_rung35ms_3x16` (line 624) is the only `from_spec` entry that omits `dm_entropy_intensive=True` and so takes the default `False`, while its meta-GGA sibling at line 669 sets it. The asymmetry is inert for a stronger reason than descriptor coverage: **the flag has no functional consumer anywhere in the package.** A quote-agnostic sweep returns 48 occurrences, and every one is a declaration (`config.py:131`), a type validation (`:208`), a `from_spec` pass-through (`:386`, `:458`), a registry literal, or a test; `DMStatisticsDescriptor` accepts no `intensive` argument at all, and the only hits outside `config.py` and the test tree are two comment strings in notebook tooling. The field survives solely in the serialized architecture record through `describe()`, so it can change a spec's SHA-256 digest and nothing else. It is a vestige of the removed `dm_entropy` feature (Section 3.2), and the naming asymmetry is an editing artifact.

3.7 The coordinate transforms

Which coordinates the MLPs read is a per-architecture, per-run switch, and the two settings are separate checkpoint families. The exchange network branches at `networks.py:274-281` and the correlation network at `:556-583`.

Under `descriptor_coordinates: dfs` (every v6 run; `model.descriptor_coordinates: dfs` in the YAML, recorded in each trained checkpoint's class record) the inputs are DFS's own, implemented at `networks.py:27-50`:

$$x_s = \left(1 - e^{-s^2}\right)\ln(s+1), \qquad x_\alpha = \ln\left(\frac{\alpha+1}{2}\right),$$
$$x_0 = \ln\left(\rho^{1/3} + 10^{-5}\right), \qquad x_1 = \ln\left[\tfrac{1}{2}\left((1+\zeta)^{4/3} + (1-\zeta)^{4/3}\right)\right].$$

These are DFS Eqs. 9, 10, 7 and 8 in turn: x_s is `_dfs_log_transform` (`:30`), x_α is `_dfs_indicator_coordinate` (`:37`), and x_0 and x_1 are inlined in the correlation network at `:574` and `:575`. Note that x_1 as written – the *logged* spin-scale factor – is DFS Eq. 8; Eq. 4 is the same factor unlogged, which is the form the legacy family feeds (below). The in-code comment at `networks.py:563` labels the logged form "eq. 4", which is the narrower citation. The offset 10^{-5} inside the density logarithm is DFS's own (`_DFS_LOG_EPS`, `:27`, from dpyscfl `net.py` line 39, `self.loge = 1e-5`). The exchange MLP reads x_s alone at the GGA level and x_s with x_α at the meta-GGA level; the correlation MLP reads (x_0, x_1, x_s) plus extras, with x_α substituted for the raw indicator column among the extras on the meta-GGA rung. The x_α substitution operates on the **raw** indicator recovered from the stored column by `networks._raw_indicator` (`:44`), which is why the invertibility measured in Section 3.5 is load-bearing.

A third documented deviation sits at the x_s line itself (`networks.py:576`), beside the Eq. 7 deviation of the legacy C-net and the Eq. 10 deviation of the legacy indicator input: the reduced gradient entering x_s is that of the **total** density, with no spin rescaling. The reference implementation's unpolarized branch divides s by $(1+\zeta)^{2/3} + (1-\zeta)^{2/3}$ before the transform, and its own source marks that line as inherited from xcdiff rather than from dpyscfl (`net.py:202`, comment "line below in xcdiff, not dpyscfl"); the repository does not apply it, and records the choice at `networks.py:565-566`. The epsilon asymmetry between the coordinates is *not* a deviation: the reference applies its `loge` offset only inside the density logarithms (`net.py:187-190`, `:226-229`) and writes the gradient coordinate as `log(descr3 + 1)` with no offset (`net.py:198`, `:204`), exactly as `_dfs_log_transform` does.

Under the legacy family (`descriptor_log_transform: true`, every run through v5) the exchange network applies the same $\{1 - e^{-x^2}\}\ln(x+1)$ form to s, but the correlation network carries a **documented deviation from DFS Eq. 7**, stated at `networks.py:540-548`: its density feature is r_s rather than DFS's $\rho^{1/3}$, and it is passed through the s-style transform (the reduced-gradient form of Eq. 9) rather than the plain logarithm Eq. 7 applies to the density variable. The deviation is recorded and not changed, since a plain-log density form would invalidate every existing checkpoint. In that family ζ enters through the bounded feature $x_1 = \frac{1}{2}[(1+\zeta)^{4/3} + (1-\zeta)^{4/3}]$ without the outer logarithm, which equals 1 at $\zeta = 0$ so that a closed-shell call sees the unpolarized input. Under `descriptor_log_transform: false` (the `deep_notransform*` entries) both variables are fed raw.

This distinction is not cosmetic for the cross-campaign comparison of Section 9: the anchored v6 `medium` cells and the unanchored v4 `deep_3x16` cells differ operatively by the anchor **plus** the legacy-to-dfs coordinate change, and by nothing else (Section 9.1).

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4. v4: unanchored pretraining to parent enhancement values

4.1 Protocol

The v4 pretraining fitted each network pair, per architecture, to the parent's enhancement factors point-wise on stored reference rows:

- **Data.** Self-consistent PBE atomic densities of the training pool's elements – for the v4 arms H, Li, C, N, O, F and Na at their ground-state spins, with He dropped as absent from the production basis (`hpcjobs/configs/dfs_step7.dfs6311_grid3_v4gga.yaml`, `pretrain.atoms`; the library default `pretrain_data_gen.DEFAULT_PRETRAIN_ATOMS` is the smaller H/He/O/N set) – on their Becke grids, with libxc's parent values stored as clipped ratio-minus-one targets. For architectures whose descriptor set is exactly the meta-GGA indicator, a synthetic regular (r_s, s, α) mesh ($7 \times 8 \times 10$ nodes; `pretrain_data_gen.MESH_RS = (0.1, 0.3, 0.7, 1.5, 3.0, 5.0, 10.0)`, `MESH_S = (0.0, 0.25, 0.5, 1.0, 1.5, 2.0, 3.0, 5.0)`, `MESH_ALPHA = (0.0, 0.1, 0.25, 0.5, 0.75, 1.0, 1.5, 2.0, 3.0, 5.0)`) was added on 2026-08-10 at a stated `MESH_WEIGHT_FRACTION = 0.3` share of the loss weight, each node realized as a physical (ρ, σ, τ) triple ($\rho = 3/(4\pi r_s^3)$, $\sigma = (2sk_F\rho)^2$, $\tau = \alpha\,\tau_{\mathrm{unif}} + \tau_W$) and evaluated through the same libxc calls as the atomic rows. The mesh exists because SCAN is three-dimensional in (r_s, s, α) and the atomic manifolds leave the α axis underdetermined: the pretrained-only meta-GGA network scored 42.65 kcal/mol on held-out reactions against SCAN's own 4.45 before the mesh, and the meta-GGA C-net was measured up to 0.457 from SCAN away from $\alpha = 1$ (HISTORY 2026-08-10; `pretrain.py` lines 1275-1277, the mesh-append rationale). Geometry-bearing descriptors (cusp, rung-3.5) cannot be defined at a geometry-free mesh node, so those meta-GGA stacks pretrained atoms-only with the caveat on record.

- **Objective.** Integration-weighted mean-squared enhancement-factor residual (quadrature weight times $|\rho\,\epsilon^{\mathrm{LDA}}|$), 2500 steps (`dfs_step7/dfs6311_grid3_v4gga/runs/run_20260810T202813Z/pretrain/deep_3x16/pretrain_metadata.json`: `pretrain_steps = 2500`, `parent_anchor` absent). No per-system energy term, no held-out validation, no acceptance gate of any kind.
- **Target policy.** Each architecture pretrained to the best functional its inputs can exactly represent: PBE for the GGA and rung-3.5 forms (faithful at ≤ 0.013 in F), SCAN for the indicator-bearing forms (HISTORY 2026-08-10, the v4 sweep definition).

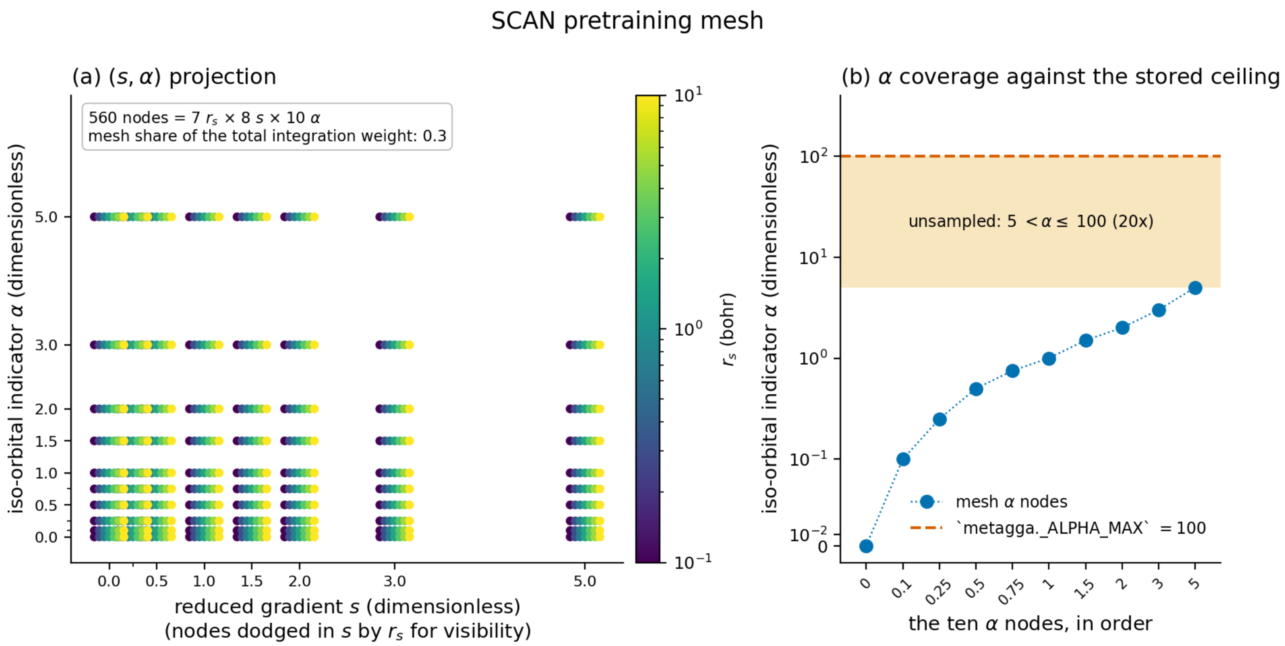

Figure 6: The synthetic SCAN pretraining mesh. Left: the (s, α) projection of all 560 nodes, coloured by r_s and dodged in s for visibility. Right: the ten α nodes against the stored column's ceiling.

The mesh is enumerated from `pretrain_data_gen.MESH_RS`, `MESH_S` and `MESH_ALPHA` themselves; `dfs_mesh.csv` records the 560 nodes ($7 \times 8 \times 10$) and their coordinates. Read the left panel for what the mesh buys: the atomic rows of a real molecule trace a one-dimensional manifold through (r_s, s, α), whereas the mesh covers the space as a product grid, so the α axis is determined independently of the density profile that generated it. Read the right panel for what it does not cover: the largest α node is 5, while the stored column runs to `_ALPHA_MAX = 100`, leaving a factor of 20 in α unsampled. **Conclusion:** the mesh regularizes the indicator dependence over the physically populated range and says nothing about the low-density tail – which is exactly why the mesh contributes nothing to the anchored pretraining floor of Section 6.5, where the entire residual is carried by capped tail rows.

4.2 What the handoff actually delivered

Measured from the pretrained checkpoints of the v4 GGA arm (`figures_dfs_step7_dfs6311_grid3_v4gga_val_best/pretrain_fx_fc_curves.csv`): the pretrained exchange curves sit max $|\Delta F_x| = 0.039$ (deep_3x16) to 0.090 (deep_rung35_3x16) from PBE over the plotted grid; the meta-GGA correlation nets sit up to 0.49–0.52 from SCAN (`figures_dfs_step7_dfs6311_grid3_{v4,v5}_val_best/pretrain_fx_fc_curves.csv`, `.../v5mgga2/pretrain_fx_fc_curves.csv`). The final point-wise exchange loss of the v4 deep_3x16 pretrain was 4.6×10^{-5} (same metadata file) – for comparison, the v6 anchored runs *start* at 2.7×10^{-32} (Section 6.5).

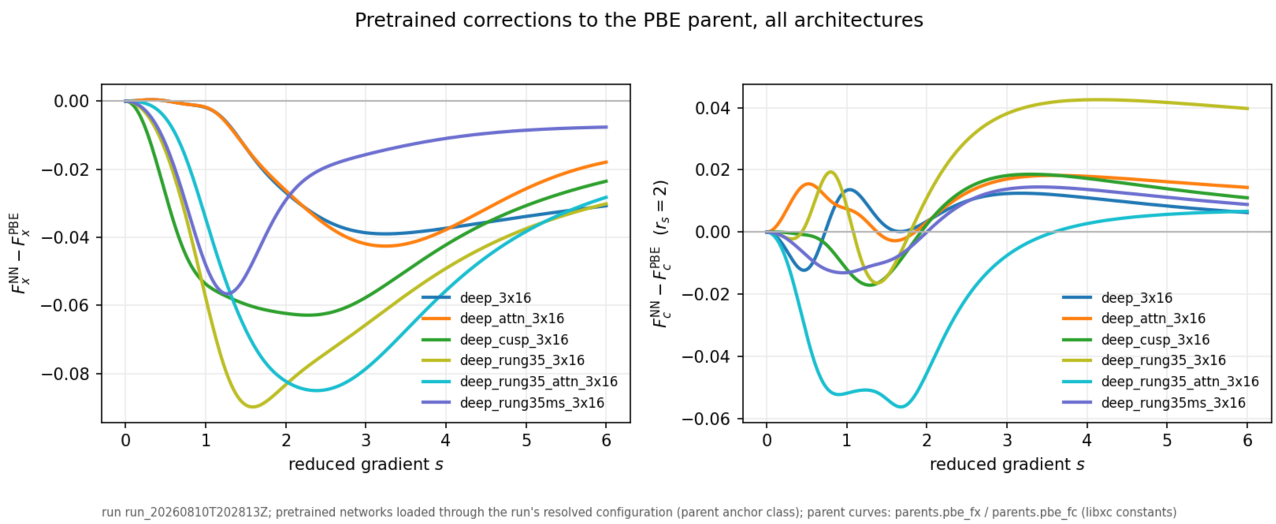

Figure 7: Unanchored (v4gga) pretrained enhancement factors against PBE: the difference panels for all six GGA architectures, F_x against s and F_c against s at three densities.

Each curve is a trained-from-flat network minus its parent, evaluated through the production model builder against `parents.pbe_fx` / `pbe_fc`. Read it as the state in which each architecture *entered* its fine-tune. The per-architecture worst deviations, recomputed from `pretrain_fx_fc_curves.csv`, are $|\Delta F_x|$ of 3.899×10^{-2} (deep_3x16), 4.259×10^{-2} (deep_attn_3x16), 5.659×10^{-2} (deep_rung35ms_3x16), 6.287×10^{-2} (deep_cusp_-3x16), 8.503×10^{-2} (deep_rung35_attn_3x16) and 8.983×10^{-2} (deep_rung35_3x16), with the correlation channel reaching 1.517×10^{-1} (deep_rung35ms_3x16). **Conclusion:** a 2500-step point-wise fit leaves the descriptor-carrying architectures two to four times further from the parent than the descriptor-free ones, and the ordering of the curve metric does not predict the energy offsets of the next paragraph. The per-architecture panels of the same set (`pretrain_fx_fc_deep_*.png` in that directory) show each pair separately; they carry no information the difference panels omit and are not reproduced here.

In energy units, the 2026-08-20 probe (frozen parent densities, production evaluation path, every value re-derived independently; HISTORY 2026-08-20, "Pretraining does not deliver the parent") measured atomization-energy offsets from the parent on H2O / N2 / CH4 at pretraining handoff of $-2.5/-4.2/-2.4$ kcal/mol (deep_3x16) and $-2.3/-4.1/-3.1$ (deep_attn_3x16) for the descriptor-free networks, against $-13.2/-4.2/-25.7$ (deep_cusp), $-13.5/-3.5/-29.1$ (deep_rung35), $-29.5/-20.4/-56.1$ (deep_rung35_attn) and $-22.0/-30.9/-42.8$ (deep_rung35ms). The separation is in the **worst system per architecture**, not uniformly system by system: each descriptor-carrying architecture entered its 200-epoch fine-tune 25.7 to 56.1 kcal/mol from its parent on its worst of the three, against 4.1–4.2 for the two descriptor-free controls. Individual systems do overlap – deep_rung35's N2 offset of 3.5 sits inside the controls' own 2.3–4.2 span – so the claim is about the worst case an architecture carries into training, on the reaction type it would be scored on. The pretraining loss did not see it – the architecture with the lowest exchange residual carried the largest offset – because a point-wise objective does not control the integral (HISTORY 2026-08-20; `SPEC_pretrain_fidelity_program.md` Section 2; `NOTES_v5_mgga_vs_scan.md:192-211`).

4.3 The two defects behind the offsets

Two distinct mechanisms, both surfaced on 2026-08-20, ended the v4/v5 record for the affected architectures (HISTORY 2026-08-20, Phase 39):

1. **The open-shell spin-scaling defect (meta-GGA architectures).** The production UKS exchange applied the Oliver-Perdew scaling by doubling ρ and quadrupling σ per spin channel but passed the *total-density* iso-orbital indicator unchanged into both channels, in the energy and in the SCF potential – so every open-shell species of every meta-GGA architecture was trained and evaluated on a functional different from its own closed-shell definition. The pretrained meta-GGA pair over-bound H2O / N2 / CH4 by 30.5 / 55.9 / 20.8 kcal/mol relative to SCAN; transforming only the indicator (α_σ from $2\rho_\sigma, 4\sigma_{\sigma\sigma}, 2\tau_\sigma$, which

libxc's spin-polarized SCAN satisfies to $< 10^{-12}$ Ha) recovered 75, 86 and 63 percent of the effect on H2O, N2 and CH4 respectively (fractions from NOTES_v5_mgga_vs_scan.md; the residual offsets -7.6 / -7.9 / -7.6 kcal/mol). The superseded two-block evaluation costs −30.1 kcal/mol on the O atom alone for SCAN exchange, and O anchors atomization energies throughout the pools. A secondary contribution came from the pretraining rows themselves: open-shell atoms stored spin-resolved SCAN targets against total-density inputs (+1.0 / +7.3 / −2.0 kcal/mol of the offsets). (HISTORY 2026-08-20; the repair is Phase 43.)

2. **The pretraining-fidelity gap (GGA descriptor carriers).** The cusp and rung-3.5 offsets are *not* spin-scaling defects – the cusp feature is pure geometry and exact under the doubled-spin substitution (Section 3.1), and the rung-3.5 occupancy has no in-domain doubled-spin evaluation. They decompose into an H-atom pretraining error shared by every cusp-carrying network (+13.7 mHa against +0.8) multiplied by the hydrogen count, plus molecular extrapolation of density-matrix features never constrained by the atoms-plus-mesh pretraining set (HISTORY 2026-08-20).

No campaign stage had ever compared a pretrained network with its parent on atoms or atomization energies; the offsets were larger than the architecture differences the campaigns existed to resolve, and this held while the campaigns ran. Every in-flight training was cancelled on 2026-08-20 and the v4/v5 descriptor-architecture record retired (HISTORY 2026-08-20).

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5. v5: per-rung self-consistent seeding

v5 changed where the truncated training SCF *starts*, not how the networks are pretrained (HISTORY Phase 37, 2026-08-14):

- **What changed.** `precompute_fixed_density_data` gained a `dm_seed` channel; meta-GGA-family architectures train and evaluate their truncated (3-cycle) SCF from a converged SCAN density matrix instead of converged PBE, matching each rung's own baseline (the DFS reference work prepares per-campaign converged PBE-or-SCAN baselines the same way). A fourth held-out evaluation channel (`eval_holdout_coldstart`: functional-free minao seed, 25 cycles, `conv_tol` 10^{-12}) was added as a trajectory diagnostic (Section 1.4). The v5 arm YAMLs are byte-derived from v4 with exactly five deltas (seed source, seed cache, coldstart flag, output root, eval wall); losses, solver, subsets, hyperparameters and the pretrain block are untouched, and the shared pretrain data file could not regenerate, so the v5 pretrained pairs replicate the v4 protocol and land at the same distances from the parent (the v4 and v5 meta-GGA pretrain curve CSVs carry identical deviations, `figures_dfs_step7_dfs6311_grid3_{v4,v5}_val_best/pretrain_fx_fc_curves.csv`).
- **What it fixed.** The density-footing asymmetry: previously every rung's truncated SCF was seeded from PBE's fixed point while SCAN comparison numbers used converged SCAN, a handicap concentrated on the beyond-GGA arms.
- **What it did not fix.** Neither defect of Section 4.3. The 2026-08-20 read of the SCAN-seeded cells against SCAN (HISTORY 2026-08-20; `notebooks/analysis/NOTES_v5_mgga_vs_scan.md`) found the cells reproducing SCAN's held-out accuracy at subset sizes 2–5 (E/E_{SCAN} 0.94–1.01) with one of 21 (leg, subset) cells beating SCAN, and identified the limiting factor as the fine-tuning regime itself: both parents' BH76 error is ~90% bias (PBE −7.5, SCAN −6.0 kcal/mol of MAE 7.7 / 6.4); fine-tuning removes the bias and injects reaction-level scatter nearly uncorrelated with the parent's residual (root-mean-square difference from SCAN 8.6–11.0 kcal/mol, correlation of the two error vectors 0.21–0.33 at subset sizes 2–5, `NOTES_v5_mgga_vs_scan.md:68-72`) – a property of the recipe, not of the seeding. The same read exposed that the cells never started at SCAN at all (Sections 4.2-4.3), which retired the v5 meta-GGA record as a quantitative result. Its trained enhancement-factor curves (`figures_dfs_step7_dfs6311_grid3_v5_val_best/trained_fx_fc_*`) document a retired record and are kept for the development narrative.

The v4 and v5 pretrained networks are the same objects – the shared pretraining data file could not regenerate between the arms – and the two directories' `pretrain_fx_fc_curves.csv` carry bit-identical deviations, which is the check that establishes it: $\max|\Delta F_c|$ reads 9.4367×10^{-2} (deep_mgga_3x16), 4.6973×10^{-2} (deep_mgga_attn_3x16) and 4.8980×10^{-1} (deep_rung35_mgga_3x16) in *both* files, with $\max|\Delta F_x|$ at 2.0487×10^{-2}, 1.7658×10^{-2} and 6.7213×10^{-2}. The v5 set's own difference panel is therefore not reproduced here: it is the same figure. **Conclusion:** the correlation channel of the rung-3.5-stacked meta-GGA form is half an enhancement factor away from SCAN at handoff – five times the worst GGA-arm exchange deviation of Section 4.2 – and no part of the v5 change touched it.

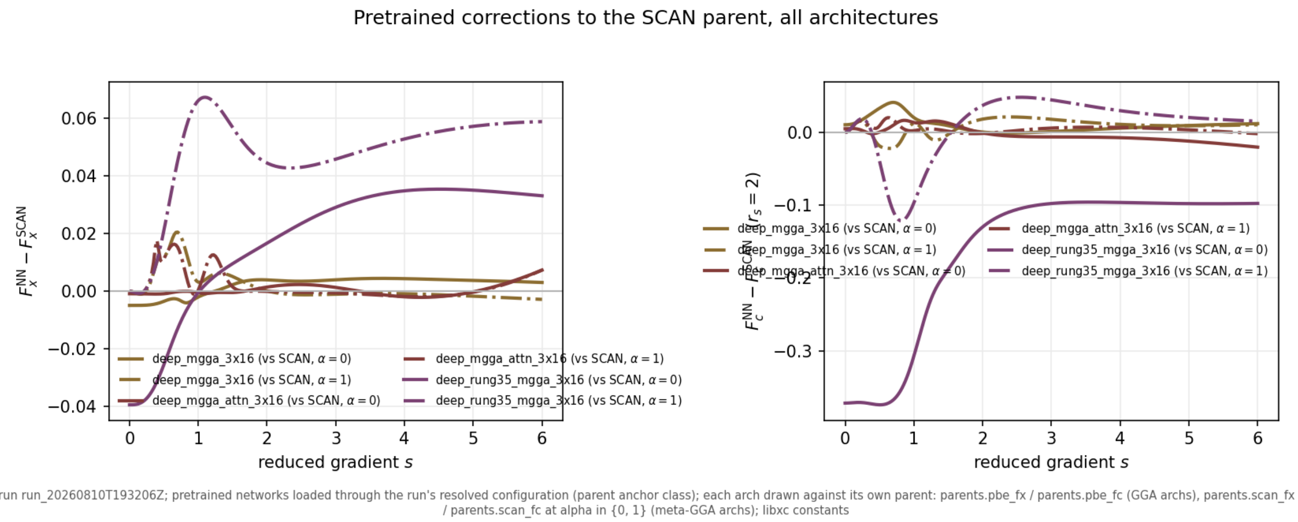

Figure 8: Unanchored meta-GGA pretrained factors against SCAN, v4/v5 arm: `deep_mgga_3x16`, `deep_mgga_attn_3x16` and `deep_rung35_mgga_3x16`, with the exchange and correlation panels at the SCAN indicator slices $\alpha = 0$ and $\alpha = 1$.

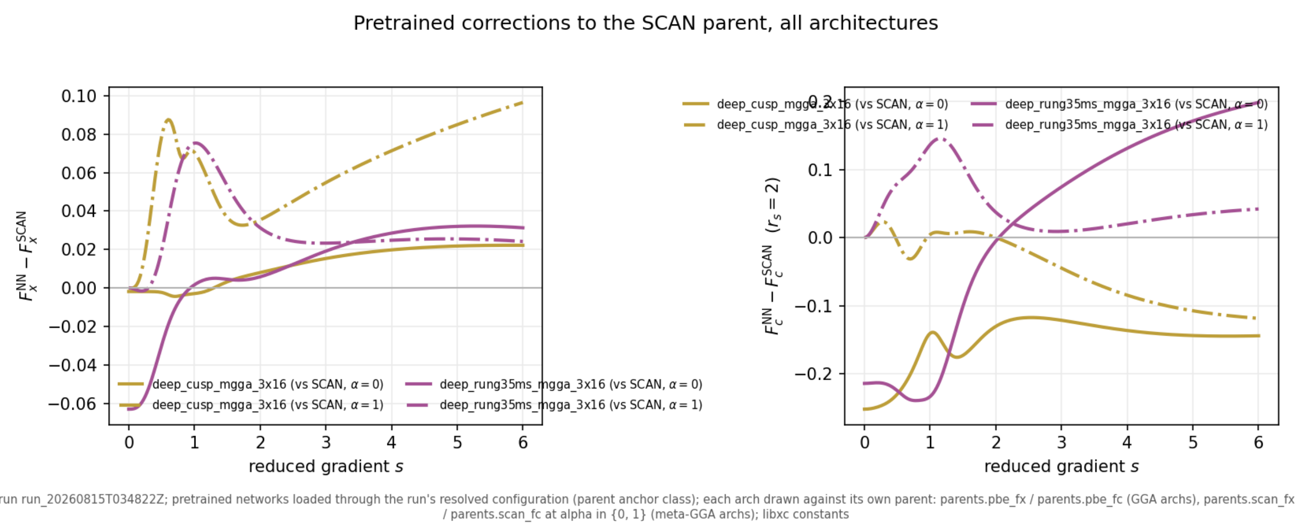

Figure 9: Unanchored meta-GGA pretrained factors, the v5mgga2 stacking arm: `deep_cusp_mgga_3x16` and `deep_rung35ms_mgga_3x16` against SCAN.

This arm never produced trained weights: no `model*.eqx` file exists anywhere under `dfs_step7/dfs6311_grid3_v5mgga2/` in the pulled tree, and the directory accordingly carries `pretrain_fx_fc_*` files and no `trained_fx_fc_*`. It is a pretraining-only record, retained because it completes the measurement of how far the legacy protocol landed from SCAN on the two most heavily stacked forms: from `pretrain_fx_fc_curves.csv`, max $|\Delta F_c|$ is 5.2084×10^{-1} (deep_cusp_mgga_3x16) and 4.8408×10^{-1} (deep_rung35ms_mgga_3x16), the worst values recorded anywhere in the campaign, with the exchange channel at 9.6451×10^{-2} and 7.5367×10^{-2}. **Conclusion:** the legacy pretraining error grew with descriptor stacking in the correlation channel specifically, and the worst case is a network that reproduces barely half of its parent's correlation enhancement before a single fine-tuning step is taken.

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6. v6: the parent-anchored pretraining protocol

v6 replaces "pretrain toward the parent and hope" with "construct the parent, then certify". Four coupled changes: the anchored network class, the exact open-shell footing with a smoothed indicator, a pretraining protocol with parent energies and validation, and a per-architecture fidelity certificate that gates every campaign stage.

6.1 The anchored construction

With `parent_anchor` set (every v6 configuration; `hpcjobs/configs/dfs_step7.dfs6311_grid3_v6*.yaml`, `model.parent_anchor: true`), the gated network term enters in the *pre-image* of the bounded map at the parent's value (`networks.AlecGGA_XNet._core`, `AlecGGA_CNet._core`):

$$F(g) = 1 + L(z_{\text{parent}} + T(g))\,, \qquad z_{\text{parent}} = \ln\left[\frac{(\Lambda - 1)\,F_{\text{parent}}}{\Lambda - F_{\text{parent}}}\right] \text{ clamped to } [-40, 40],$$

with F_{parent} evaluated on the row's own physical inputs by the JAX parent implementations (`parents.pbe_fx` / `pbe_fc` / `scan_fx` / `scan_fc`, every constant the value libxc 7.0.0 carries) and the pre-image `parents.lob_preimage` (`parents.py:633`). The clamp at $z_{\max} = 40$ binds only where the parent approaches a bound of $(0, \Lambda)$: within $\Lambda(\Lambda - 1)e^{-40}$ of the ceiling (8.5×10^{-18} at $\Lambda = 2$, 6.2×10^{-18} at 1.804, 8.7×10^{-19} at 1.174) and within $\Lambda e^{-40}/(\Lambda - 1)$ of the floor (8.5×10^{-18} at $\Lambda = 2$, 9.53×10^{-18} at 1.804, 2.87×10^{-17} at 1.174); both logarithm arguments are floored at the smallest normal float so a parent at or past a bound by round-off clamps instead of returning NaN (`parents.lob_preimage` docstring, as corrected). The construction guarantees $F \in (0, \Lambda)$ for every $T(g)$ and returns F_{parent} exactly at $T(g) = 0$.

That identity is exact to one unit in the last place, and the qualification matters where the ceiling is 1.174. Measured through the committed classes: $F(0) = 1$ bitwise at $\Lambda = 1.804$ and $\Lambda = 2.0$, and $F(0) = 0.9999999999999999 = 1 - \varepsilon/2$ at $\Lambda = 1.174$, where ε is the double-precision epsilon – a single rounding in $\exp(\ln(\Lambda - 1))$. The full pre-image round trip $F \to z \to F$, taken through `_AlecLOB` in the production form $F = 1 + L(z)$ over $F \in [10^{-3}, \Lambda - 10^{-3}]$, closes to a few units in the last place, and the measured maximum is a sampling statement: a uniform 2×10^6-point grid reads 4.44×10^{-16} at $\Lambda = 1.804$ (worst near $F = 1.32$), 2.78×10^{-16} at 1.174 and 2.22×10^{-16} at 2.0, while dense sampling near the worst region at 1.174 reaches 3.33×10^{-16} – per-point round-off extremes grow with sampling density, so these are observed lower bounds on the true worst, all of order one to two ulp of F. Part of the residual is the $1 + L$ addition itself: comparing $L(z)$ against $F - 1$ directly, without re-adding the one, tightens 1.174 to 2.22×10^{-16} on the same grid. These are the floors every "reproduces its parent to 2.2×10^{-16}" statement below rests on, not fitted agreements.

`create_network_pair` forces `zero_init_final_layer` for an anchored pair, so both networks equal the parent at initialization by construction. Measured: the anchored meta-GGA architectures return $F_x =$ `scan_fx` within 2.8×10^{-16} and $F_c =$ `scan_fc` within 2.2×10^{-16} on 31,550 exchange and 15,790 correlation rows of OH and H2O (HISTORY 2026-08-25); a freshly built anchored pair reproduces the parent curves under 10^{-10} (F_x) and 10^{-8} (F_c) on the figure grid where an unanchored build differs by more than 10^{-2} (HISTORY 2026-08-30, the pretrained-figure identity pins). An anchored correlation network must be polarization-aware (ζ-blind construction refused by name, `networks.py:462`), because the parent correlation is divided by the model's ζ-dependent PW92 baseline whose curvature pole is the subject of Section 2's third figure.

The v6 runs use the DFS descriptor coordinates (`model.descriptor_coordinates: dfs`; the trained-model class records state it, e.g. `dfs_step7/dfs6311_grid3_v6g1_size/runs/run_20260827T163330Z/checkpoints/spec_-0009/model_val_best.eqx.class.json`): the exchange MLP reads $x_s = (1 - e^{-s^2}) \ln(s+1)$ of the doubled channel and, on the meta-GGA rung, $x_\alpha = \ln((\alpha+1)/2)$ of the raw indicator – the network inputs of DFS Eqs. 9-10, defined in Section 3.7.

6.2 Exact open-shell footing and the smoothed indicator

Every density-matrix-derived descriptor is now evaluated on the symmetric doubled density of its own spin channel, $\mathrm{diag}(P_\sigma, P_\sigma)$ – in the precompute, the SCF loop of both solver backends, the energy, the potential, the losses and the pretraining rows – so the open-shell exchange is the same functional the closed-shell path defines. With libxc SCAN in place of the network, the three-block energy reproduces PySCF's spin-polarized SCAN exchange to 1.8×10^{-15} Ha on O and OH, and the assembled potential is the finite-difference derivative of the energy to 1.0×10^{-10} Ha worst case; closed-shell paths are bitwise unchanged (HISTORY Phase 43, 2026-08-20 to 2026-08-24).

The meta-GGA indicator (`metagga.compute_alpha`, `metagga.py:163`) is $\alpha = (\tau - \tau_W)/\tau_{\mathrm{unif}}$, stored as $\min(p(\alpha_{\mathrm{raw}}), 100)$ with p the smooth positive part

$$p(x) = \tfrac{1}{2}\left(x + \sqrt{x^2 + w^2}\right), \qquad w = 10^{-5},$$

which replaced a hard lower clip whose derivative was discontinuous exactly on the manifold every one-electron spin channel occupies (`metagga.smooth_positive_part`, `:142`; `_ALPHA_SMOOTHING_WIDTH = 1e-5` at `:104`, `_ALPHA_MAX = 100` at `:61`). The width is anchored to measurement, not chosen: it must dominate the floating-point residue of $\tau - \tau_W$ on one-orbital channels (worst on-domain residue 1.3–3.7×10^{-6}, margin 2.7–7.7x), and its SCAN energy cost is 1.17×10^{-7} Ha on the H atom, linear in the width, 8.5×10^3 below the certificate's free-atom tolerance (`metagga.py` width commentary; HISTORY 2026-08-24). What the clip cost is on record: with it, the beta-channel feature-response term of Li's Fock matrix moved by 0.93 Ha under a 10^{-14} relative change of the density matrix, and H's by 4.2×10^{-3} Ha; with the smooth positive part the same probe moves H's by 3.6×10^{-12} Ha (`metagga.py:181-199`). The definition string `metagga.ALPHA_DEFINITION` (`:112`) is part of the pretraining-data manifest identity, so a file computed under another definition is stale by construction.

6.3 The pretraining set and objective

- **Systems.** The DFS pretraining set (eight free atoms and 22 G2/97 molecules, committed as package data at `xcquinox/alec/data/dfs_pretrain_set.json`, its bytes pinned by the SHA-256 in `xcquinox/alec/tests/test_dfs_pretrain_set.py`, line 189) at the rung's level – the meta-GGA variant of the DFS protocol drops H2 and N2 (`cluster/fidelity.dfs_level_for_parent`) – merged with every free atom of the BH76 / W4-11 pools at the pools' own charges and spins, de-duplicated by geometry (`pretrain.resolve_pretrain_systems`; the G1 run records 38 systems and 1.39M exchange / 1.21M correlation rows, `run_20260827T163330Z/pretrain/medium/pretrain_metadata.json`). This set is distinct from the 26-point *training* pool of Section 1.2 and from the DFS Letter's own 21-molecule training set; conflating them is easy and the three are separate objects (`xcquinox/alec/dfs_pretrain_set.py:1` against `dfs_pool.py:1-12`). Rows are built on the *parent's* reference density – PBE or SCAN by rung (`inputs.parent_density: auto`) – under the training orientation lock (3×10^{-5}), with reference SCFs converged through a two-stage DIIS-then-second-order ladder that refuses to write an unconverged record (HISTORY Phase 40; the second stage starts from the best point of the DIIS trajectory since 2026-08-30, which is what let the SCAN reference of the Li atom at 6-311++G(3df,2pd) converge and the meta-GGA datagen complete, HISTORY 2026-08-31).
- **Exchange footing.** `exchange_footing: spin_channel`: each open-shell exchange row is posed per spin channel at $(2\rho_\sigma, 4\sigma_{\sigma\sigma})$ with descriptor blocks of the doubled density, reproducing libxc spin-polarized PBE exchange to 3.6×10^{-12} Ha on O (HISTORY Phase 43). The historical `total` footing is what every run through v5 used.
- **Objective.** The point-wise integration-weighted residual, plus an optional per-system energy term $w_E \frac{1}{N_{\mathrm{sys}}} \sum_s \left(\sum_{i\in s} w_i\, \epsilon_i^{\mathrm{LDA}} F_i^{\mathrm{NN}} - E_s\right)^2$ holding the network to the parent's own energies on the same quadrature (`pretrain._PretrainLoss`; targets close on PySCF to 4.9×10^{-13}–4.5×10^{-11} Ha, HISTORY

Phase 42). Under the anchor the shipped campaigns run `energy_term_weight: 0.0` as the exact statement of the objective – both terms are zero to round-off at initialization – while the term remains *measured* on the saved network (recorded `energy_term_max_abs_dE_mHa` 2.6e-3 for G1 medium, 3.5e-3 for deep_mgga). For an *unanchored* run the same configuration is refused, because the point-wise objective was measured unable to deliver the parent (2.3–56.1 kcal/mol of atomization offset, `SPEC_pretrain_fidelity_program.md` Section 2) and an energy-weight sweep then measured that no weight closes the gap either (`SPEC_parent_anchor.md` Section 2; both cited in the G1 YAML's objective block).

- **Validation.** A seeded 20% of the multi-nucleus systems is withheld (never an atom, since every pool atom anchors an atomization energy), scored every 50 steps with patience 10, and the best-validation network is the one written; a run whose validation values are all non-finite raises instead of returning the untrained network (HISTORY Phase 42). The G1 `medium` record: held-out systems CO2, HCl, C4H6, C3H8; 2500 steps run; best at step 2500 (`pretrain_metadata.json`, `validation` block).
- **Mesh.** The synthetic (r_s, s, α) mesh is kept, as a regularizer at the 0.3 share, for meta-GGA architectures whose descriptors a mesh node can define: in the v6 meta-GGA group, `deep_mgga_3x16` and `deep_mgga_attn_3x16` carry it (`pretrain_mesh: true`, `mesh_loss_share_x` 0.3) while the geometry-bearing stacks do not; the GGA groups run without it (`pretrain_metadata.json` of the respective runs).

6.4 The per-architecture fidelity certificate

`xcquinox/alec/cluster/fidelity.py` accepts a pretrained pair only when it reproduces its parent in energy units, on the parent's own self-consistent density, through the production energy path, at the run's SCF identity:

$$\Delta E_{xc} = E_{xc}^{\mathrm{NN}}[\rho_{\mathrm{parent}}] - E_{xc}^{\mathrm{parent}}[\rho_{\mathrm{parent}}], \qquad \Delta AE(\mathrm{mol}) = \Delta E_{xc}(\mathrm{mol}) - \sum_{\mathrm{atoms}} n_{\mathrm{atom}}\, \Delta E_{xc}(\mathrm{atom}),$$

with PASS requiring $\max_{\mathrm{atoms}} |\Delta E_{xc}| \leq$ `tol_atom` $= 1.0$ mHa AND $\max |\Delta AE| \leq$ `tol_AE` $= 1.0$ kcal/mol (the program's binding decision; a tolerance above 2.0 requires a written override reason). The oracle set is every free atom of the BH76 / W4-11 pools, the DFS pretraining set at the rung's level, H2O / N2 / CH4 at the *pool* geometries for every rung (so the three headline offsets are one physical quantity across architectures), and ground-state Li and Na – 39 systems for a GGA rung (16 atoms + 23 atomizations), 38 for a meta-GGA one (16 + 22: the meta-GGA DFS protocol drops H2 and N2 while the fixed oracle pool restores N2) (`fidelity.build_oracle_set`, `dfs_level_for_parent`; the pulled certificates' `summary` blocks). The parent's E_{xc} per system is computed three independent ways (point-wise libxc on the stored grid, PySCF numint on a fresh grid, the reference SCF's own accumulated value), with any pairwise disagreement above 10^{-6} Ha a named failure (worst recorded spread at the production identity: `max_parent_grid_diff_Ha` 3.04×10^{-9} over the four G1 certificates, 2.28×10^{-9} over the five meta-GGA ones); degenerate open-p-shell atoms are measured on orientation-locked references; a non-finite value is a named failure, never a silent pass. With the parent itself presented behind the model interface the certificate is an identity to 3.6×10^{-15} Ha (PBE) and 2.0×10^{-10} Ha (SCAN) on the O atom (HISTORY Phase 40).

The verdict, every per-system number, the run identity, the SHA-256 digests of the two network files and the installed code version are written to `<run>/pretrain/<arch>/fidelity_certificate.json`. Enforcement has two layers: the on-node gates (pretrain task exit code, train task, preflight, in-process model builder) honour a recorded waiver for workflow smoke tests; the record layers (`validate_run`, the cross-arm merge, the figure suite) require PASS unconditionally, so a waived run can never become a quantitative result (HISTORY Phase 40).

The gate has fired on real submissions. The size ladder was first submitted unanchored: its certificates read FAIL at 0.79–3.01 mHa / 1.63–11.3 kcal/mol (worst: shallow), against the 1.0 / 1.0 gate (`dfs_step7/dfs6311_grid3_v6g1_size/runs/run_20260827T123919Z/pretrain/*/fidelity_certificate.json`, `enforced: true`, `parent_anchor: false`; first-step pretrain losses 0.008–0.012). The anchored resubmission passes at $7.2–8.5 \times 10^{-4}$ mHa / $1.9–3.9 \times 10^{-3}$ kcal/mol (`run_20260827T163330Z`, same files). The DM-descriptor group shows the same pair for two of its three architectures: unanchored FAIL at 5.76 mHa / 11.7 kcal/mol (deep_combined) and 5.87 / 11.4 (deep_dm), anchored PASS at $1.2–5.1 \times 10^{-3}$ mHa for all three (`dfs_step7/dfs6311_grid3_v6g3_dm/runs/run_20260827T124112Z` and `...T163335Z`; deep_combined_attn has no unanchored control – that run wrote no certificate for it). The five meta-GGA family architectures pretrained to PASS certificates at production identity (6-311++G(3df,2pd) / grid 3, lock 3×10^{-5}), worst `max_atom` 5.15×10^{-3} mHa (deep_mgga_attn) and worst `max_dAE` 2.5×10^{-3} kcal/mol (deep_mgga), over 38 systems each (`dfs_step7/dfs6311_grid3_v6g2_families_mgga/runs/run_20260831T011905Z/pretrain/*/fidelity_-certificate.json`; HISTORY 2026-08-31). Twelve anchored PASS certificates exist across the pulled v6 groups (4 G1 + 5 meta-GGA + 3 DM).

6.5 The measured pretraining floors, and the SCAN-parent floor derivation

Because the anchored start *is* the parent, the first-step point-wise pretraining loss is a direct measurement of how exactly the stored targets and the anchored parent agree:

group (parent)	architectures	first-step exchange loss
G1 size ladder (PBE)	shallow, shallow_attn, medium, medium_attn	2.72×10^{-32} (all four)
G2 meta-GGA (SCAN), mesh-carrying	deep_mgga, deep_mgga_attn	3.02×10^{-14}
G2 meta-GGA (SCAN), mesh-free	deep_cusp_mgga, deep_rung35_mgga, deep_rung35ms_mgga	4.31×10^{-14}

(`.../pretrain/<arch>/losses_x.npy`, step 1, of `run_20260827T163330Z` and `run_20260831T011905Z`.) The PBE-parent floor is the double-precision identity: residuals of order 10^{-16} in F, squared. The 18-orders-of-magnitude gap to the SCAN-parent floor is carried entirely by the indicator ceiling, `metagga._ALPHA_MAX` = 100 – not by the smoothing (HISTORY 2026-08-31 (erratum), which supersedes the first recorded derivation). The stored column is $\min(p(\alpha_{\mathrm{raw}}), 100)$, and the anchored network recovers the raw indicator from it exactly below the ceiling (`networks._raw_indicator` / `metagga.invert_smooth_positive_part`; Section 3.5's right-hand panel is that inversion measured): a stored $p(0) = w/2 = 5 \times 10^{-6}$ inverts to $\alpha = 0.0$ to round-off, the end-to-end anchored exchange MSE on the committed mesh block is 7.62×10^{-32}, and the H atom – one-orbital on every row, the column at its smoothed floor everywhere – prices at 2.85×10^{-32} (HISTORY 2026-08-31 (erratum)). At the ceiling the inversion cannot act: the column reads 100 while the exact indicator on the capped rows is unrecoverable. The capped population is the low-density tail: on the O atom 2492 rows are capped, with α_{exact} spanning $\sim 1.0 \times 10^{2}$ to $\sim 7.0 \times 10^{6}$ (median 555) at ρ from 10^{-10} to 8.6×10^{-5} (median 6.1×10^{-8}), 90% of the capped weight sitting at $\alpha = 108$–2475.

This is the price of the ceiling in the quantity the pretraining loss measures. Read it as follows: a grid row whose exact indicator exceeds 100 is evaluated at 100, and the vertical axis is the resulting error in the parent's own exchange enhancement factor. The residual is zero by construction at the cap and rises to a saturation value as $\alpha \to \infty$; `alpha_ceiling.csv` gives that saturation as 1.7365×10^{-3} at $s = 0$, 1.2421×10^{-3} at $s = 1$ and 1.0008×10^{-3} at $s = 4$, against $F_x(s = 0, \alpha = 100) = 0.78598$ – so at most about 0.2 percent relative. SCAN's switching function has nearly but not exactly saturated at the cap: per capped row of the O atom the median $|\Delta F|$ against the exact-τ libxc target is 2.55×10^{-4} and the worst is 5.70×10^{-4}. **Conclusion:** the ceiling is not a bug and not a precision regression; it is a stated, bounded truncation whose F-space cost is at most about 0.2 percent relative (1.74×10^{-3} absolute in F_x, at $s = 0$) and whose energy cost the certificate gates at 194 times inside the atomic tolerance.

The capped rows carry 100.0% of the weighted exchange MSE on the O atom (1.2593×10^{-13} on one converged reference path, 1.2190×10^{-13} on a second, the capped share 1.000000 on both) and on H2O (3.07×10^{-14}), the uncapped remainder pricing at 2.7×10^{-29} and 0.0 (HISTORY 2026-08-31 (erratum, as amended)). The synthetic mesh contributes nothing ($\leq 6 \times 10^{-29}$; its α nodes stop at 5, below the ceiling – `pretrain_data_gen.py` line 1070, `MESH_ALPHA`, and the right-hand panel of the mesh figure in Section 4.1), so the mesh-carrying floor is the mesh-free floor scaled by exactly the atomic share of the loss weight: $3.0167 \times 10^{-14}/4.3096 \times 10^{-14} = 0.7000000000000004$ (the ratio from `losses_x.npy` step 1 of the two file variants). A hypothetical in which the parent reads the smoothed column directly, the inversion removed, prices the mesh block at 1.90×10^{-14}, worst $|\Delta F| = 5.6 \times 10^{-7}$ at $(r_s = 0.1, s = 0, \alpha = 0)$ (the coordinates as the superseded derivation recorded them) – numerically adjacent to the measured floors, which is what that derivation mistook for the mechanism – but it was never the run's code path (HISTORY 2026-08-31 (erratum)). `parents.scan_fx` itself agrees with libxc SCAN to 4.9×10^{-15} max over $\rho \in [10^{-6}, 10^{2}]$, $s \leq 8$, $\alpha \leq 10$ (HISTORY 2026-08-31), so the floor is not an implementation error of the parent. Nothing is repaired because nothing is broken: the ceiling is the recorded energy-faithfulness bound on an indicator whose raw value grows without bound on the low-density tail (`metagga.py`, HISTORY Phase 17) – it zeroes the indicator's derivative only at capped points, and the separate tail-gradient freeze that once accompanied it was removed on 2026-08-06 for misreporting the derivative of an energy ingredient (`metagga.py`

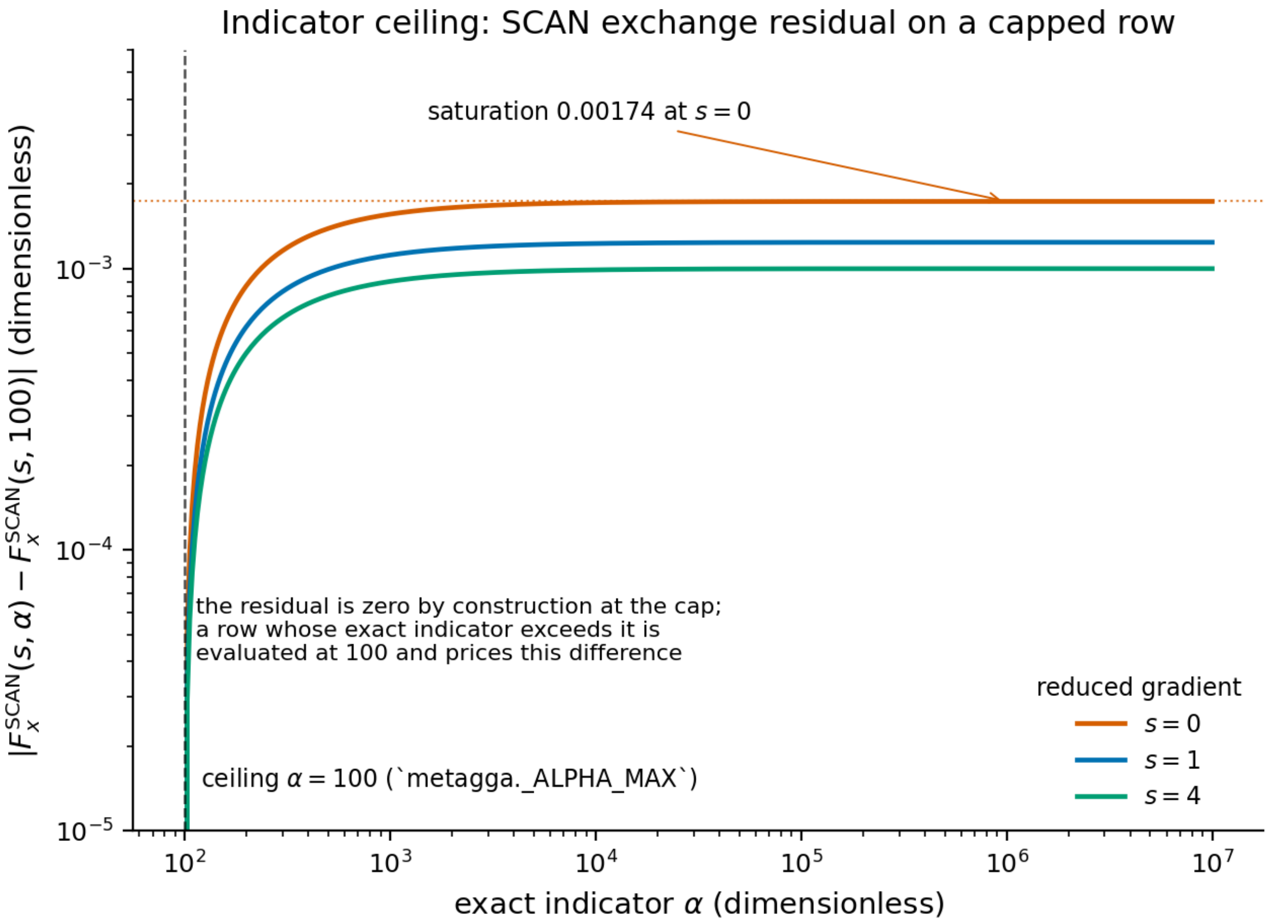

Figure 10: Indicator ceiling: the SCAN exchange residual on a capped row. Each curve is $|F_x^{\mathrm{SCAN}}(s,\alpha) - F_x^{\mathrm{SCAN}}(s,100)|$ against the exact indicator, at three reduced gradients.

lines 55-60; HISTORY 2026-08-06). Its energy consequences – 8.8×10^{-8} Ha on the N atom's exchange for the ceiling and 1.2×10^{-7} Ha on the H atom for the smoothing – are stated floors of the SCAN oracle four orders under the 1.0 mHa free-atom gate (1.14×10^4x and 8.3×10^3x; `SPEC_parent_anchor.md` Section 3.1), with the ceiling's F_x effect 1.8×10^{-3} absolute in F_x on the capped rows (about 0.2 percent relative, as priced above); and the certificates gate the whole start's consequence at 194x inside the atomic tolerance (worst meta-GGA `max_atom` 5.15×10^{-3} mHa, deep_mgga_attn, against 1.0 mHa). The corrected derivation is recorded so the 10^{-14} start reads as the priced consequence of a documented bound rather than as a precision regression.

6.6 Pretrained enhancement factors

The pretrained curves are now a published, per-generation artifact (`notebooks/analysis/pretrain_fx_fc.py`, loading through the production certified-model builder; baselines are `parents.pbe_fx` / `pbe_fc` – the anchor's own parent at libxc constants, because a rounded-constant analytic helper differs by up to 4.553×10^{-6} and would read as a spurious learned correction under the anchor; HISTORY 2026-08-30):

- v6 G1 (anchored, PBE parent), after the 2500-step pretrain: $\max|\Delta F_x| = 8.7 \times 10^{-7}$ (shallow), 4.0×10^{-6} (shallow_attn), 4.2×10^{-6} (medium_attn), 9.2×10^{-6} (medium); the correlation channel spans 1.1×10^{-6} (medium) to 2.2×10^{-5} (shallow_attn) (`figures_dfs_step7_dfs6311_grid3_v6g1_size_val_best/pretrain_fx_fc_curves.csv`).
- v6 G2 meta-GGA (anchored, SCAN parent): all five architectures within 8.1×10^{-7} to 1.3×10^{-5} of SCAN over both channels (`figures_dfs_step7_dfs6311_grid3_v6g2_families_mgga/pretrain_fx_fc_curves.csv`; per-channel worst values from the same file: F_x 8.1×10^{-7} (deep_cusp_mgga) to 1.3×10^{-5}

(deep_mgga), F_c 5.4×10^{-6} to 1.1×10^{-5}).

- The legacy (unanchored) pretrains, same figure pipeline: $\max|\Delta F_x|$ 0.039–0.090 (v4gga GGA arms), meta-GGA F_c up to 0.49–0.52 (`figures_dfs_step7_dfs6311_grid3_{v4gga,v4,v5}_val_best/pretrain_-fx_fc_curves.csv`, `.../v5mgga2/...`).

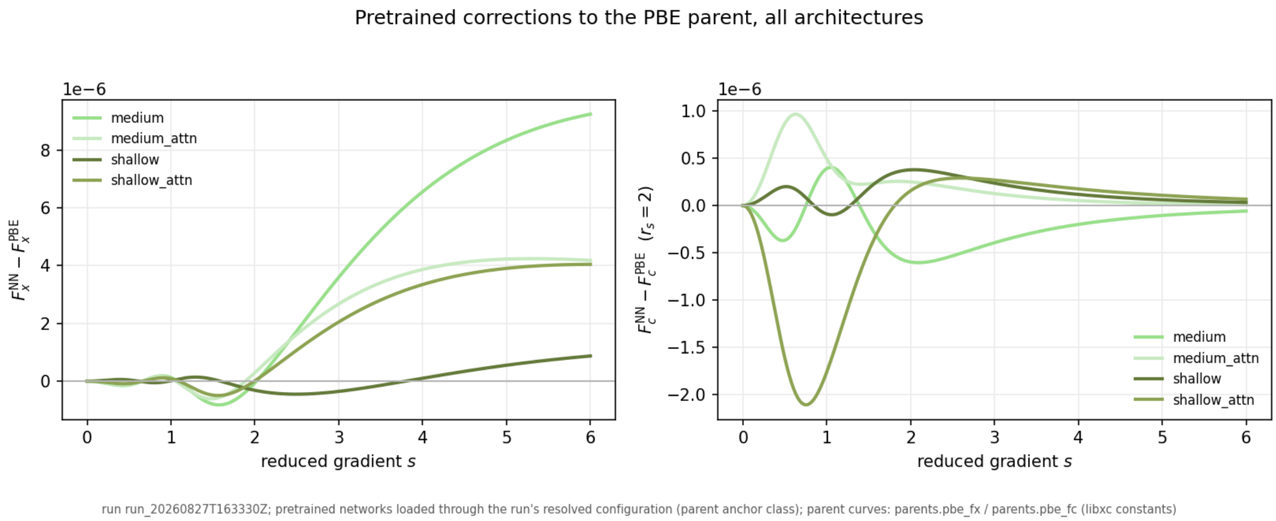

Figure 11: Anchored (v6 G1) pretrained enhancement factors against PBE, all four size-ladder architectures. Note the vertical scale.

This is the same figure, from the same module, as the v4gga panel of Section 4.2 – and the only thing that has changed is the axis. The worst deviation anywhere on this figure is 2.1957×10^{-5} (`shallow_attn`, correlation channel, from `pretrain_fx_fc_curves.csv`), against 1.517×10^{-1} on the v4gga panel: a factor of about 7000. **Conclusion:** whatever the anchored networks later learn, they do not begin by having to relearn their parent, and the residual structure visible here is the 2500-step pretrain's own drift away from an exact start rather than an approach toward one.

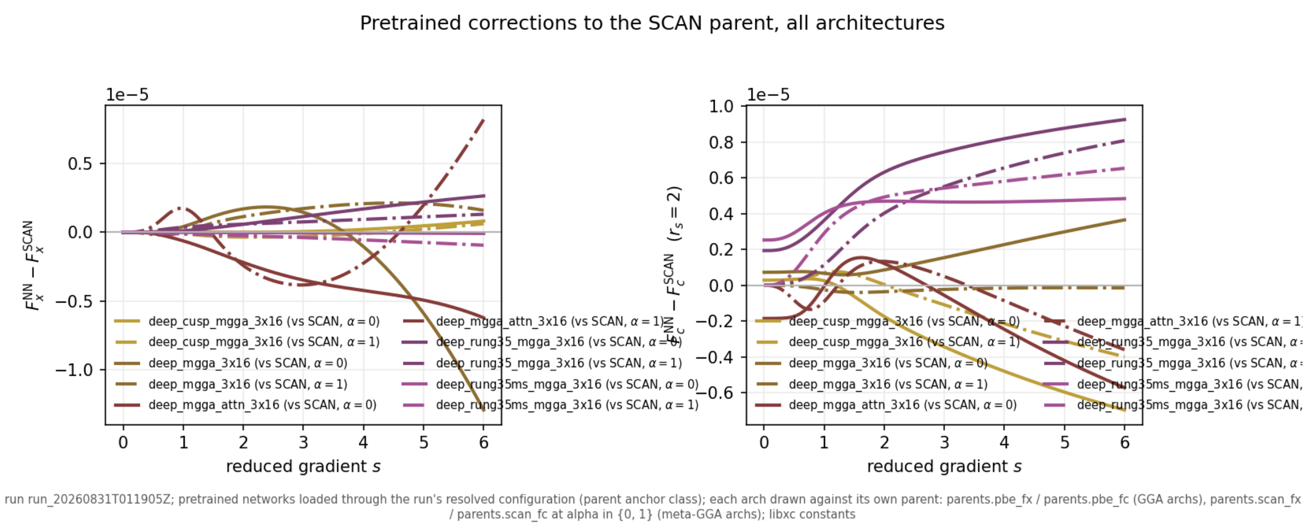

Figure 12: Anchored (v6 G2) meta-GGA pretrained factors against SCAN, all five family architectures, at the indicator slices $\alpha = 0$ and $\alpha = 1$.

The meta-GGA counterpart, against the harder parent. From `pretrain_fx_fc_curves.csv` the worst exchange deviations are 8.094×10^{-7} (deep_cusp_mgga_3x16) to 1.2901×10^{-5} (deep_mgga_3x16) and the worst correlation deviations 5.385×10^{-6} to 1.1197×10^{-5}. Compare the v5mgga2 panel of Section 5, whose correlation channel reaches 5.2×10^{-1} on the same two stacked architectures. **Conclusion:** the anchor removes the descriptor-stacking dependence of the pretraining error entirely – the five architectures now agree with SCAN to within a factor of 16 of each other, where the legacy protocol spread them over two orders of magnitude and made the most information-rich architectures the least faithful.

The anchor bought pretraining fidelity of four orders of magnitude in the curve metric (HISTORY 2026-08-31).

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7. The comparison figure sets

Per-generation enhancement-factor sets, all rendered by the same two modules (`notebooks/analysis/pretrain_-fx_fc.py`, `trained_fx_fc.py`) against the same parent baselines:

set	architectures	contents
`figures_dfs_step7_dfs6311_grid3_v3_val_-best`	deep, deep__attn, deep__cusp (3x16)	pretrain + trained
`figures_dfs_step7_dfs6311_grid3_v4gga_-val_best`	six GGA forms	pretrain + trained
`figures_dfs_step7_dfs6311_grid3_v4_val_-best`	deep__mgga, deep__mgga__attn (+ rung35__mgga pretrain)	pretrain + trained (retired record)
`figures_dfs_step7_dfs6311_grid3_v5_val_-best`	deep__mgga (+ trio pretrain)	pretrain + trained (retired record)
`figures_dfs_step7_dfs6311_grid3_v5mgga2`	deep__cusp__mgga, deep__rung35ms__mgga	pretrain only
`figures_dfs_step7_dfs6311_grid3_v6g1_size`	shallow(__attn), medium(__attn)	pretrain + `anchored_vs_unanchored_fx_fc.png`
`figures_dfs_step7_dfs6311_grid3_v6g1_-size_val_best`	shallow(__attn), medium(__attn)	pretrain + trained + `anchored_vs_unanchored_fx_fc.png`
`figures_dfs_step7_dfs6311_grid3_v6g2_-families_mgga`	five meta-GGA family forms	pretrain only (training pending)

Two of these are omitted from the embedded figures above on measured grounds. The v3 pretrained deltas are the v4gga ones for the three shared architectures (deep_3x16 F_x 3.8989×10^{-2} against 3.8990×10^{-2}; deep_cusp_3x16 6.2870×10^{-2} in both files), and the v5 pretrained deltas are bit-identical to v4's, as Section 5 shows. Neither adds a curve the embedded panels do not already carry.

`anchored_vs_unanchored_fx_fc.png` (in both v6g1 directories) overlays the anchored G1 corrections on the unanchored v3 and v4gga records. Its provenance chain to the published v4 merged sets (`figures_dfs6311_v4_-merged_val_best{,_gga}`) was verified by execution: symlink resolution into `run_20260810T202813Z`, byte-equal evaluation files, weights older than their evaluations in all 54 specs, single-generation job records (HISTORY 2026-08-31). The long-form data behind every curve statement below is `trained_fx_fc_curves.csv` / `pretrain_-fx_fc_curves.csv` in the respective directories, with the evaluation channel recorded per row (`eval_channel`, validation-best with a labelled final-checkpoint fallback).

These two are the trained counterparts of the pretrained panels of Sections 4.2 and 6.6, and are read as a pair. Both draw the *end* of training rather than the start, each architecture at the largest cell it has completed, so the vertical scales are comparable (order 10^{-1}) even though the starting scales differed by four orders of magnitude. From `trained_fx_fc_curves.csv`, the exchange corrections at subset size 18 span $[-0.0441, +0.0908]$ for unanchored deep_3x16 and $[-0.0226, +0.1616]$ for anchored medium; the correlation corrections at $r_s = 2$ span $[0, +0.916]$ and $[0, +0.431]$ respectively. **Conclusion:** the anchored networks do move, and by an amount comparable to the unanchored ones in exchange; the divergence is in correlation, where the anchored maximum is less than half the unanchored one, and the next figure isolates why.

This figure is the document's central comparison, and it is rebuilt by a committed module (`anchored_vs_-unanchored_fx_fc.py`) from the committed long-form curve CSVs rather than by an ad hoc script, with a footer that states its own coverage from the rows it read. The subset size 18 representative is the largest cell every drawn generation has reached, enforced by an internal consistency test; the footer currently reports v6 trained coverage of 27 cells (medium 10, medium_attn 10, shallow 7), all on the validation-best channel, which matches the tally of Section 9.2 exactly.

Read the top row first: the two unanchored pretrained curves lie on one another at 3.9×10^{-2} in F_x and 1.4×10^{-2} in F_c – the same protocol and seed – while the two anchored ones are invisible at this scale, at 9.2×10^{-6} and 6.0×10^{-7} (all from `anchored_vs_unanchored_fx_fc.csv`). Then read the bottom row. The **exchange** panel shows one family: every optimized curve has a bond-region dip and a positive bump peaking between $s = 2.35$

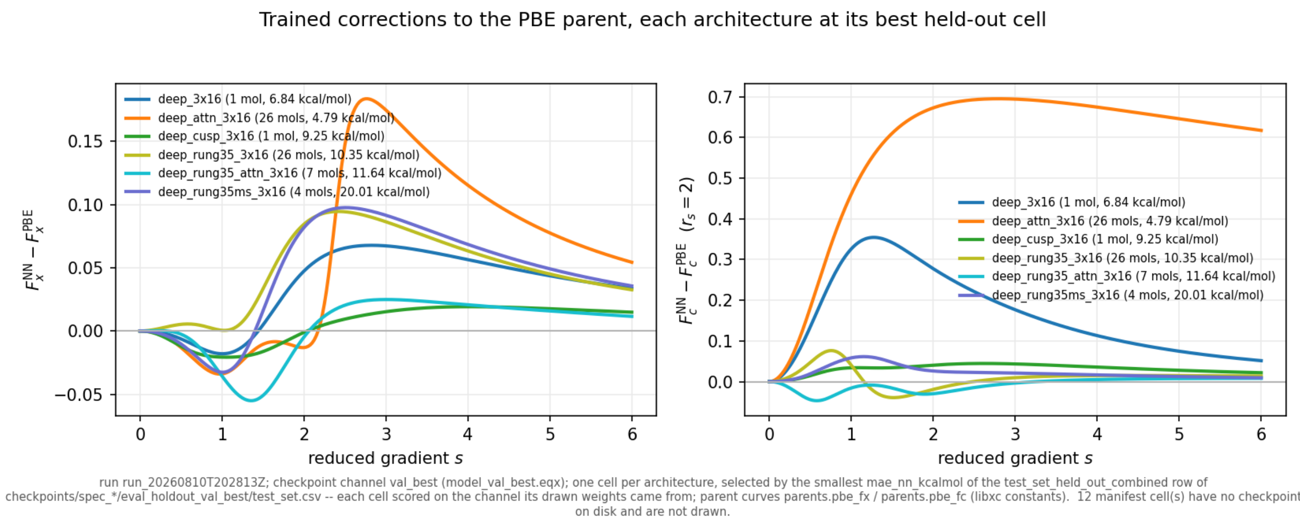


Figure 13: Optimized (validation-best) enhancement-factor differences from PBE, unanchored v4gga, one cell per architecture at its largest completed subset size.

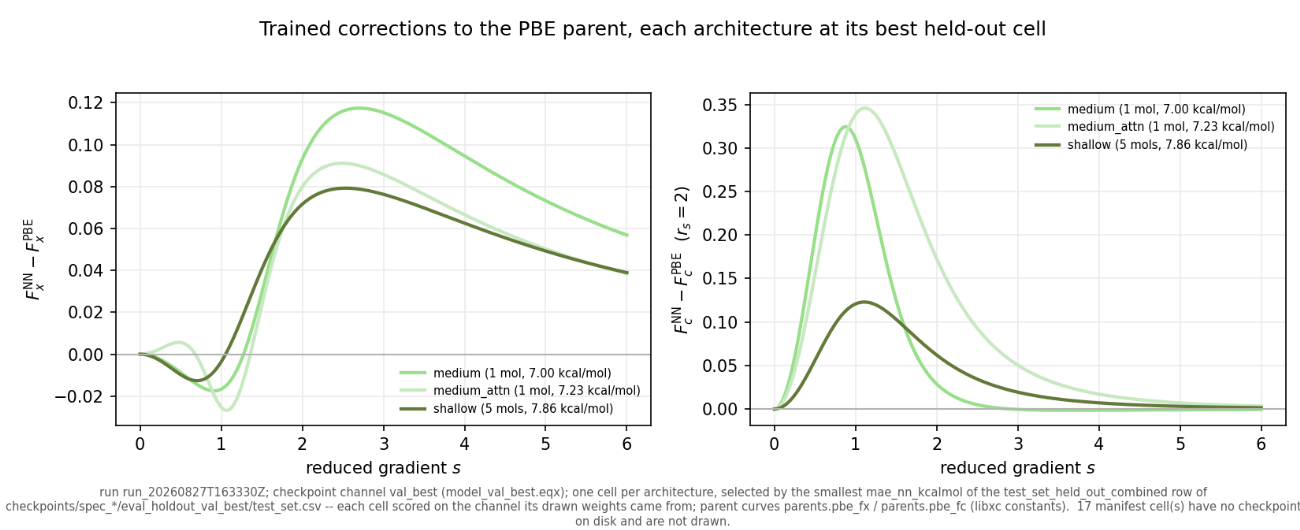
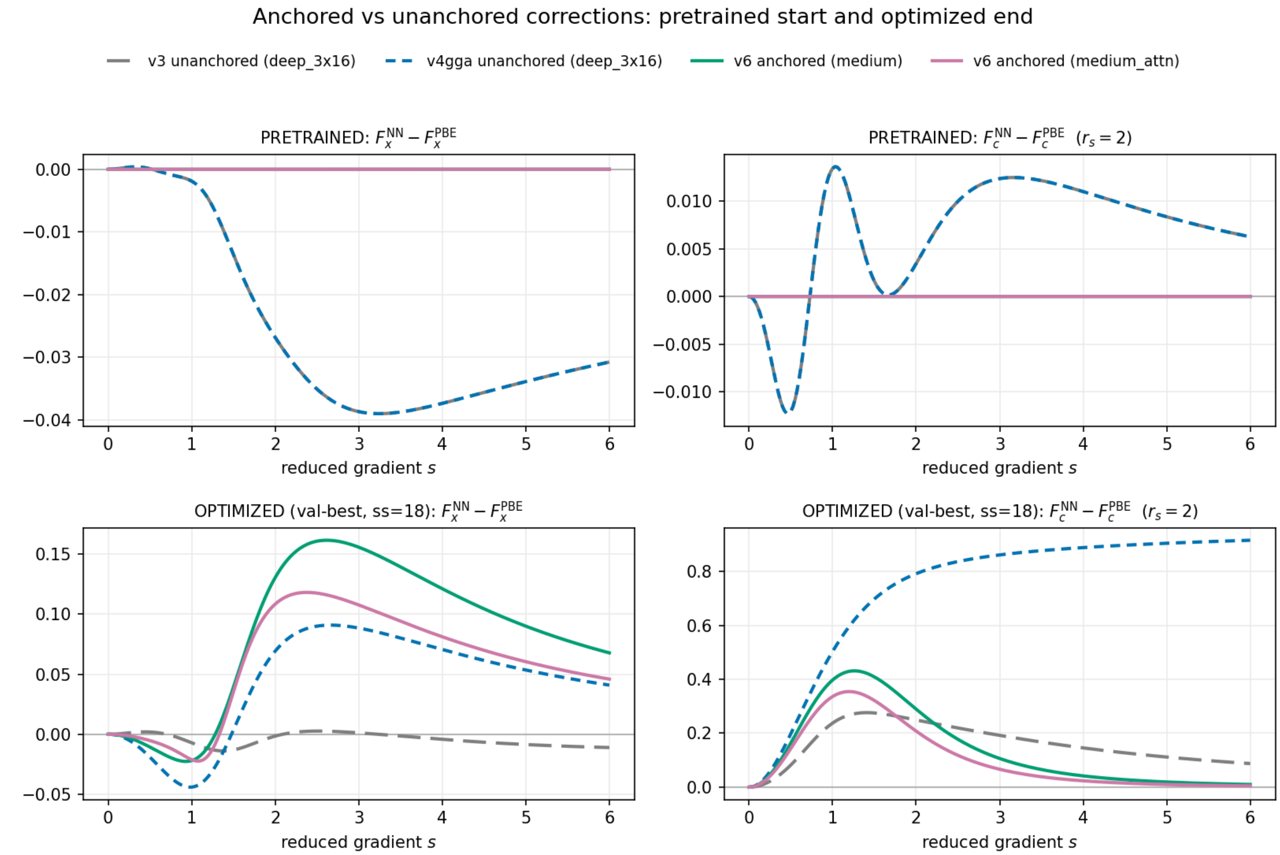


Figure 14: Optimized (validation-best) enhancement-factor differences from PBE, anchored v6 G1, one cell per architecture at its largest completed subset size.



Sources (committed long-form curves, one row per grid point): pretrain_fx_fc_curves.csv and trained_fx_fc_curves.csv of figures_dfs_step7_dfs6311_grid3_v3_val_best, figures_dfs_step7_dfs6311_grid3_v4gga_val_best, figures_dfs_step7_dfs6311_grid3_v6g1_size_val_best. Bottom row: the ss=18 cell of each series on the val_best channel. v6 trained coverage in figures_dfs_step7_dfs6311_grid3_v6g1_size_val_best: 27 cells (medium 10, medium_attn 10, shallow 7), every row on the val_best channel. Zero line = parents.pbe_fx / parents.pbe_fc (libxc constants), the anchor's own parent. Pretrained max|delta| over the drawn channels: anchored 6.0e-07 to 9.2e-06 (below the line width at this scale), unanchored 1.4e-02 to 3.9e-02. The unanchored pretrains agree to 9.9e-06, so their top-row curves lie on top of one another (separate dash periods).

Figure 15: Anchored against unanchored corrections, pretrained start and optimized end. Rows: pretrained (top) and optimized at subset size 18 (bottom). Columns: F_x against s (left) and F_c at $r_s = 2$ (right). Four series: v3 and v4gga `deep_3x16` unanchored (dashed, different dash periods), v6 `medium` and `medium_attn` anchored (solid).

and $s = 2.7$, of height $+0.0908$ (unanchored deep_3x16), $+0.1181$ (medium_attn) and $+0.1616$ (medium), with the v3 cell an order of magnitude smaller at $+0.0026$. The **correlation** panel shows two families: the unanchored correction grows monotonically into large s, reaching $+0.916$ at $s = 6$, while both anchored corrections peak near $s = 1.2$ ($+0.4314$ and $+0.3545$) and then collapse, to $+0.0099$ and $+0.0048$ at $s = 6$. **Conclusion:** training discovers the same exchange physics from either start, and does not discover the same correlation physics. Section 8.2 identifies the mechanism as a property of the anchored parameterization rather than of the optimizer.

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8. Measured pros and cons of the anchor

8.1 What the anchor buys

1. **Exact parent start.** First-step losses at the identity floor (Section 6.5); pretrained curves 10^{-7}–10^{-5} from the parent against 0.04–0.5 for the legacy protocol (Section 6.6); the certificate passes at initialization by construction, and a 2500-step pretrain cannot lose it (all twelve pulled anchored certificates PASS, Section 6.4).
2. **Certificate-gated fidelity.** The silent handoff class of Section 4.2 – worst-system offsets of 25.7 to 56.1 kcal/mol per descriptor-carrying architecture, against 4.1–4.2 for the descriptor-free controls – is structurally excluded: an architecture that does not reproduce its parent cannot reach the train stage (measured FAIL-then-PASS pairs on the same architectures, Section 6.4).

3. **A convergent exchange family.** Where anchored and unanchored campaigns can be compared at the same capacity, the *trained* exchange corrections agree in form: a bond-region dip near $s = 0.7$–1.1 (-0.016 to -0.044 over the four cells) and a positive bump peaking near $s = 2.4$–2.7, of height $+0.087$ (shallow, ss=7, its largest completed cell), $+0.091$ (unanchored deep_3x16, ss=18), $+0.118$ (medium_attn, ss=18), $+0.162$ (medium, ss=18); at $s = 3$ the band is $+0.083$ to $+0.156$ (recomputed from `trained_fx_fc_curves.csv` of `v4gga_val_best` and `v6g1_size_val_best`; HISTORY 2026-08-31 records the band as $+0.07$ to $+0.16$ near $s = 3$ at 18-cell coverage, where `shallow` reached only ss=5 and read $+0.079$; the unanchored attention twin peaks higher, $+0.210$). Training discovers the same exchange physics from either start – the optimized networks move off the parent by up to 0.16 in F_x (HISTORY 2026-08-31).

8.2 The cost: pre-image sensitivity suppression

The anchored correction enters through the bounded map at the parent's pre-image, so its parameter sensitivity carries the factor

$$L'(z_{\text{parent}}) = F_{\text{parent}} \left(1 - \frac{F_{\text{parent}}}{\Lambda}\right),$$

which vanishes as the parent approaches either bound of $(0, \Lambda)$.

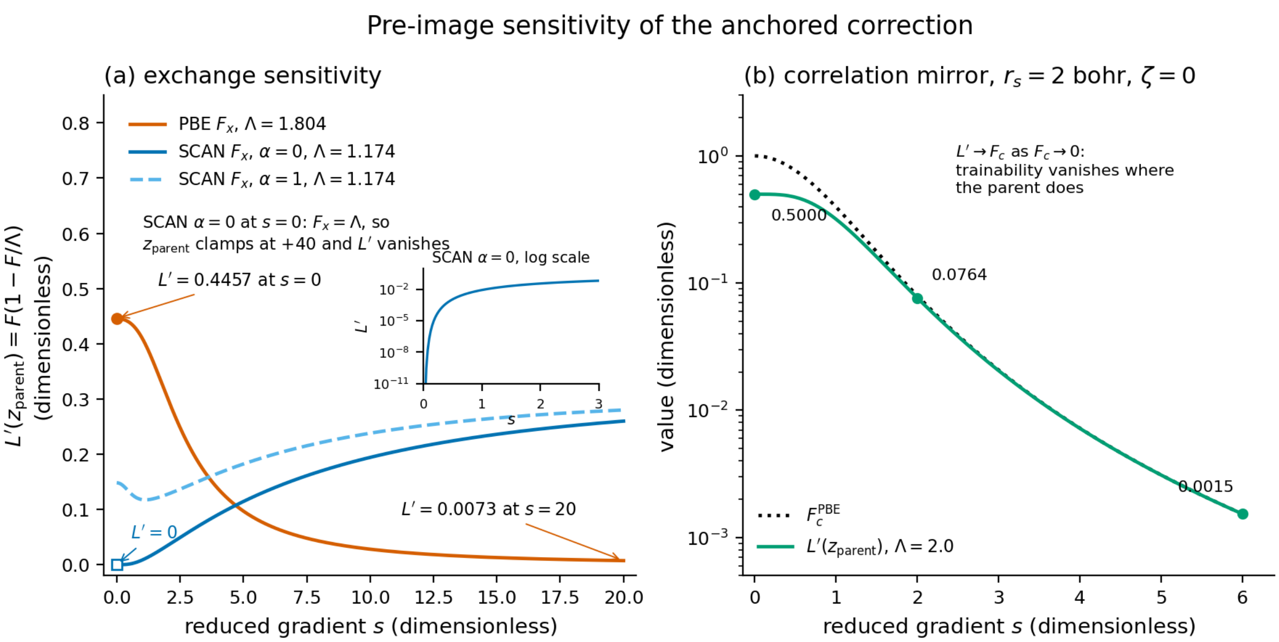

Figure 16: Pre-image sensitivity of the anchored correction. Left: exchange, $L'(z_{\text{parent}})$ against s for the PBE parent at $\Lambda = 1.804$ and the SCAN parent at $\Lambda = 1.174$ for both indicator slices, with the SCAN $\alpha = 0$ case on a log inset. Right: the correlation mirror at $r_s = 2$, $\zeta = 0$, showing L' tracking F_c down to zero.

Every value on this figure is obtained by differentiating the committed `_AlecLOB` class with `jax.grad` at the pre-image `parents.lob_preimage` returns, so it is the gradient prefactor the optimizer actually sees. Read the left panel as the exchange story: `preimage_sensitivity.csv` gives $L' = 0.4456762749$ at $s = 0$ for the PBE parent, falling to 0.0072655692 by $s = 20$ – a factor of 61 across the plotted range, but never zero. The SCAN $\alpha = 0$ curve is the degenerate case: its parent *sits on* the ceiling at $s = 0$, so the pre-image clamps at $+40$ and L' is **exactly 0.0** there, rising to 0.2599 by $s = 20$; the anchored meta-GGA exchange network is untrainable at the single-orbital small-gradient point, by construction. Read the right panel as the correlation mirror: as $F_c \to 0$ the factor $L' \to F_c$, and the two run together down the plot – 0.5000000000 at $s = 0$, 0.0764106655 at $s = 2$, 0.0015311495 at $s = 6$, against F_c values of 0.9999979292, 0.0795769076 and 0.0015323235. **Conclusion:** the anchored parameterization suppresses trainability exactly where the parent vanishes, and for PBE correlation that region is the large-gradient tail – a factor of 327 less gradient at $s = 6$ than at $s = 0$. HISTORY 2026-08-31 records the exchange endpoints rounded, as 0.45 and 0.007.

The measured consequence, on ss=18 validation-best cells at $r_s = 2$ (`trained_fx_fc_curves.csv` of `v4gga_val_best` and `v6g1_size_val_best`; HISTORY 2026-08-31): the *unanchored* v4gga deep_3x16 correlation correction grows into large s – +0.79 at $s = 2$ to +0.92 at $s = 6$ – built entirely by training from a pretrained start that sits within about ±0.014 of the parent at $r_s = 2$ (`v4gga_val_best/pretrain_fx_fc_curves.csv`); the *anchored* medium correction collapses past $s = 2$: +0.29 at $s = 2$ to +0.010 at $s = 6$ (medium_attn: +0.21 → +0.005). Both anchored cells still build sizable corrections where the pre-image leaves trainability (peaks +0.43 and +0.35 near $s = 1.2$).

8.3 The BH76 barrier bias

The unanchored campaigns used precisely that large-s correlation freedom to remove the parent's systematic barrier bias. PBE's BH76 error is almost pure bias (−6.6 to −7.5 kcal/mol mean signed error on the two slices below; |bias|/MAE = 0.97 for PBE and 0.93 for SCAN on the full v5 held-out pool, `NOTES_v5_mgga_vs_scan.md:79-89`). Measured mean signed BH76 errors on the strict held-out slice, validation-best channel (recomputed from `per_reaction.json` of the respective specs):

cell	mean signed BH76 error (NN)	PBE same slice	slice
unanchored deep_3x16, ss=12 (`merged_v4_arms/checkpoints/spec_0007`)	−0.20	−6.62	50 reactions
anchored medium, ss=12 (`v6g1_size .../spec_0007`)	−7.75	−7.47	61 reactions
anchored medium, ss=18	−4.41	−7.47	61 reactions
anchored medium_attn, ss=12	−0.81	−7.47	61 reactions

The recorded comparison pair is the first two rows (HISTORY 2026-08-31): the unanchored network removed the bias, the anchored medium cell kept it essentially at the parent's value. The last two rows, added here from the same files, show the suppression is not absolute – the anchored attention twin at the same subset size removed most of the bias (through other channels; its correlation still collapses at large s, Section 8.2) while paying for it in scatter (BH76 MAE 10.47 against PBE's 7.73, Section 9.2). Whether anchored cells can reproduce the unanchored barrier improvement is the campaign's live question (Section 12).

Unanchored cons, for the same ledger: no parent fidelity at handoff (the worst-system offsets of 25.7 to 56.1 kcal/mol per descriptor-carrying architecture in Section 4.2, against 4.1–4.2 for the controls, and invisible to the point-wise loss); every correction, physical or not, must be created by training from a flat start, so the pretrained state carries none of the parent's structure into the fine-tune beyond the ≤ 0.09 curve proximity; and there is no gate – the v4 record carried its offsets into production for the whole life of both running arms.

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9. Current numbers: completed v6 GGA cells against the corrected v4/v5 GGA record

9.1 What is being compared

- **v6 G1** (`dfs6311_grid3_v6g1_size`, run `run_20260827T163330Z`): the anchored size ladder – shallow, shallow_attn (2x8) and medium, medium_attn (3x16, the production width). Three registry fields separate `medium` from `deep_3x16` (`descriptor_log_transform`, `zero_init_final_layer`, `dm_entropy_intensive`), and under the v6 model block the two named flags are inert for this descriptor-free pair specifically – `parent_anchor` forces the zero-initialized final layer in `networks.create_network_pair`, and `descriptor_coordinates: dfs` takes the branch of both `_core` paths that never applies the log transform to the MLP coordinates (Section 3.7); the scope matters, because for a cusp-carrying architecture the flag stays live under every coordinate set through `ArchitectureConfig.materialize_descriptors`, which injects it into the cusp descriptor's own logarithm (a measured 0.51 shift of the bounded c_1 column; Section 3.1) – while the third field has no functional consumer in the package at all and reaches only the spec digest (Section 3.6); the operative cross-campaign difference between the anchored medium cells and the unanchored v4 deep_3x16 cells is therefore the anchor plus the legacy-to-dfs coordinate change alone (HISTORY 2026-08-31 (erratum), superseding the 2026-08-30 transform/initialization reading). 44 cells (4 architectures x 11 subset sizes), evaluated at the validation-best checkpoint over strict held-out slices.

- **v4gga merged validation-best** (`merged_v4_arms`, evaluations symlinked into `run_20260810T202813Z`): the corrected v4 GGA record – after the 2026-08-13 strict-holdout validity pass (the 2026-08-13 entries under the HISTORY Phase 36 header), the 2026-08-18 full-slice comparator anchors, and the NaN-species backfill (HISTORY Phase 38) – 54 of 66 cells with validation-best evaluations (11 each for deep_3x16, deep_attn, deep_cusp, deep_rung35; 7 rung35ms; 3 rung35_attn). The v5 arms contribute no GGA cells (their GGA rows are the v4 rows by design, Section 5).

The strict slices are cell-matched but not identical across campaigns: each cell is scored against its own run's held-out slice with the cell's own PBE anchor (over the 54 v4 GGA cells: BH76 43–50, W4-11 97–113, combined 145–163 reactions; over the 27 v6 cells: 61 / 111–120 / 172–181; per-spec `test_set.csv` `n_reactions` and `per_reaction.json`). Beats verdicts below are therefore within-cell (NN MAE < the same slice's PBE MAE), never cross-campaign row-for-row.

9.2 v6 G1 at current coverage: 27 of 44 cells

From the 27 `eval_holdout_val_best/test_set.csv` files present (medium 10 of 11, medium_attn 10 of 11, shallow 7 of 11, shallow_attn 0; the medium ss=26 cell failed on the open NaN-gradient defect, HISTORY 2026-08-31), all values kcal/mol, NN / cell-matched PBE:

arch	ss=1	2	3	4	5	6	7	12	15	18
W4-11 medium	6.48/13.57	11.38/13.47	10.86/13.47	11.38/13.55	9.19/13.47	10.32/13.58	8.50/13.48	12.25/13.41	9.54/13.36	7.84/13.08
medium_-attn	5.99/13.57	11.03/13.47	9.04/13.47	11.56/13.55	9.54/13.47	9.50/13.58	7.04/13.48	9.13/13.41	8.06/13.36	6.83/13.08
shallow	10.48/13.57	10.16/13.47	8.74/13.47	8.97/13.55	8.56/13.47	8.67/13.58	10.41/13.48	–	–	–
BH76 medium	8.03/7.73	9.71/7.73	9.96/7.73	10.47/7.73	6.51/7.73	10.22/7.73	6.51/7.73	10.24/7.73	8.13/7.73	6.92/7.73
medium_-attn	9.66/7.73	7.45/7.73	9.01/7.73	11.30/7.73	8.72/7.73	8.95/7.73	7.61/7.73	10.47/7.73	10.75/7.73	10.57/7.73
shallow	7.34/7.73	7.84/7.73	7.84/7.73	6.46/7.73	6.47/7.73	7.22/7.73	8.67/7.73	–	–	–
combined medium	7.00/11.59	10.82/11.53	10.56/11.53	11.07/11.58	8.29/11.53	10.29/11.60	7.82/11.52	11.55/11.43	9.05/11.38	7.51/11.18
medium_-attn	7.23/11.59	9.83/11.53	9.03/11.53	11.47/11.58	9.27/11.53	9.32/11.60	7.23/11.52	9.60/11.43	9.00/11.38	8.16/11.18
shallow	9.42/11.59	9.38/11.53	8.43/11.53	8.12/11.58	7.86/11.53	8.18/11.60	9.82/11.52	–	–	–

Headline, at 27-cell coverage:

- **W4-11: beaten in 27 of 27 cells** – NN 5.99–12.25 against cell-matched PBE 13.08–13.58, a ratio of 0.441 (medium_attn, ss=1) to 0.914 (medium, ss=12).
- **Combined pool: 26 of 27** – the one miss is medium ss=12 (11.55 against 11.43, ratio 1.011).
- **BH76: 9 of 27**, best 6.46 (shallow, ss=4) against 7.73; the other beats are medium ss=5/7/18 (6.51, 6.51, 6.92), medium_attn ss=2/7 (7.45, 7.61) and shallow ss=1/5/6 (7.34, 6.47, 7.22). The worst cell is medium_attn ss=4 at 11.30, a ratio of 1.463.

The 27-cell state supersedes the 25-cell one published on 2026-08-31 in two independent ways. Two `shallow` cells completed (ss=6 and ss=7), and seven cells were repaired for a reference defect (Section 9.3) – `medium_attn` ss=15 and ss=18 and all five then-completed `shallow` cells – which moved their W4-11 and combined rows and their PBE anchors, though not their BH76 rows. The verdict counts moved from 25/25, 24/25 and 8/25 to 27/27, 26/27 and 9/27; the one combined-pool miss is the same cell in both. The first published set, 18 cells, read 18 of 18, 17 of 18 and 5 of 18 (HISTORY 2026-08-31).

This is the combined-pool row of the table above, drawn. Read the left panel against the dashed PBE line: every one of the 27 anchored cells except `medium` at ss=12 sits below it (26 of 27, derived from the `test_set.csv` files above; combined NN 7.00–11.55 against PBE 11.18–11.60). Read the *shape* rather than the level, because there is one training run per point and no replicates (Section 1.1): the curves are not monotone in subset size, the best combined cells being at the smallest sizes (`medium` ss=1 at 7.00) and at the largest (`medium` ss=18 at 7.51), with a pronounced worsening through ss=2 to ss=6. Read the right panel as the control: in-sample error falls with subset size as it must, so the non-monotone held-out curve is a generalization property and not a fitting failure.

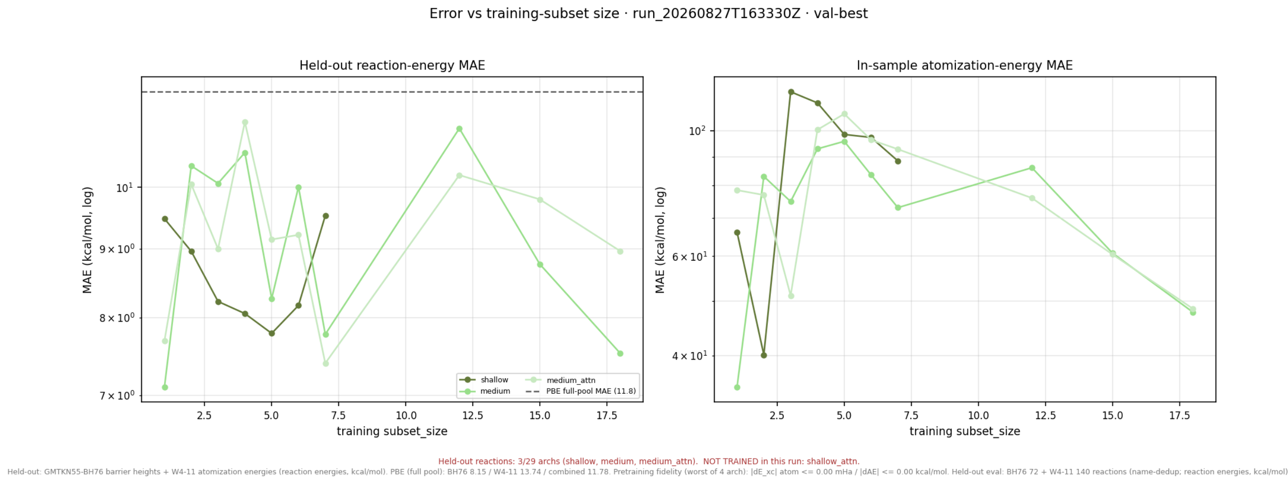

Figure 17: Learning curves: held-out reaction-energy MAE against training subset size, one line per architecture, with the full-pool PBE reference dashed. Left panel held-out, right panel in-sample.

Conclusion: at this coverage the anchored ladder shows no capacity effect that the noise of single runs would not explain, and the 2x8 `shallow` architecture is not distinguishable from the 3x16 pair on the combined pool.

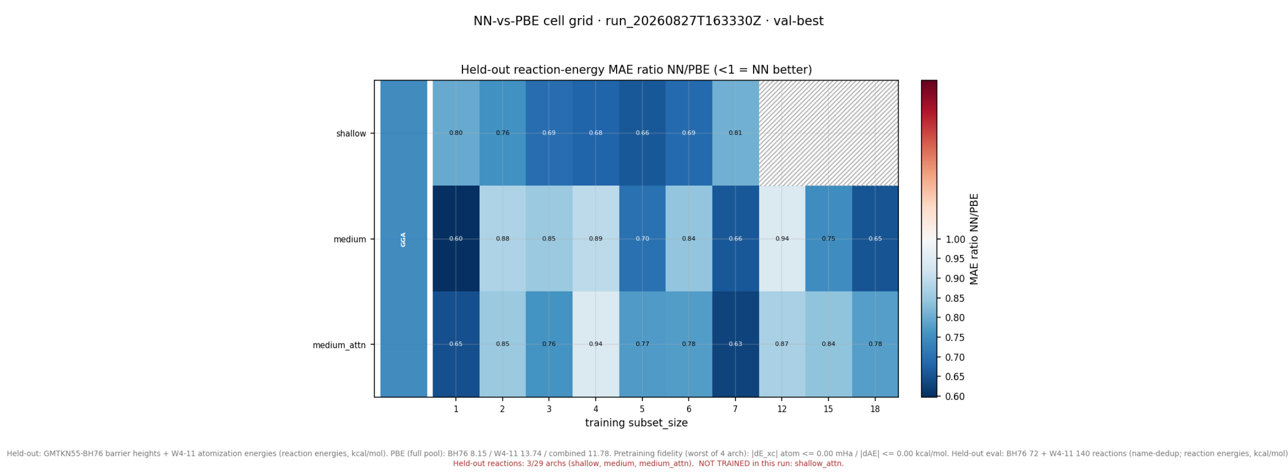

Figure 18: NN-against-PBE cell grid: the ratio of held-out reaction-energy MAE, architecture by subset size, diverging about 1.0. Blue is better than PBE; missing cells are hatched.

The same data as a grid, and the most compact statement of coverage. Each cell's PBE mean absolute error is computed on exactly the reactions its network was scored on, so every ratio is like-for-like. Over the 27 present cells the combined-pool ratio runs from 0.604 (`medium`, ss=1) to 1.011 (`medium`, ss=12); the hatched cells are the 17 not yet complete – all eleven `shallow_attn`, four `shallow` sizes, and the ss=26 cells of both 3x16 architectures. **Conclusion:** the grid reads almost uniformly blue, which is the headline result of this section, and its hatching is the honest statement of what remains: no `shallow_attn` cell exists at all, so the size ladder's attention comparison is currently one-sided.

Of the eight parity variants the suite renders, this is the one embedded, because the document's central finding is a *class* asymmetry – W4-11 beaten everywhere, BH76 rarely – and this is the only variant that separates the classes with per-class axis limits. The others are marginals over the same points: `ablation_parity.png` and `ablation_ae_parity.png` collapse to two panels (all reactions, and W4-11 alone), `ablation_parity_marginal_2x2.png` splits arch against subset without splitting class, and `ablation_parity_arch_cols.png`, `_facet_subset`, `_grid_by_subset` and `_errbars_by_subset` re-facet the identical scatter by one further variable each. None carries a class distinction this one lacks.

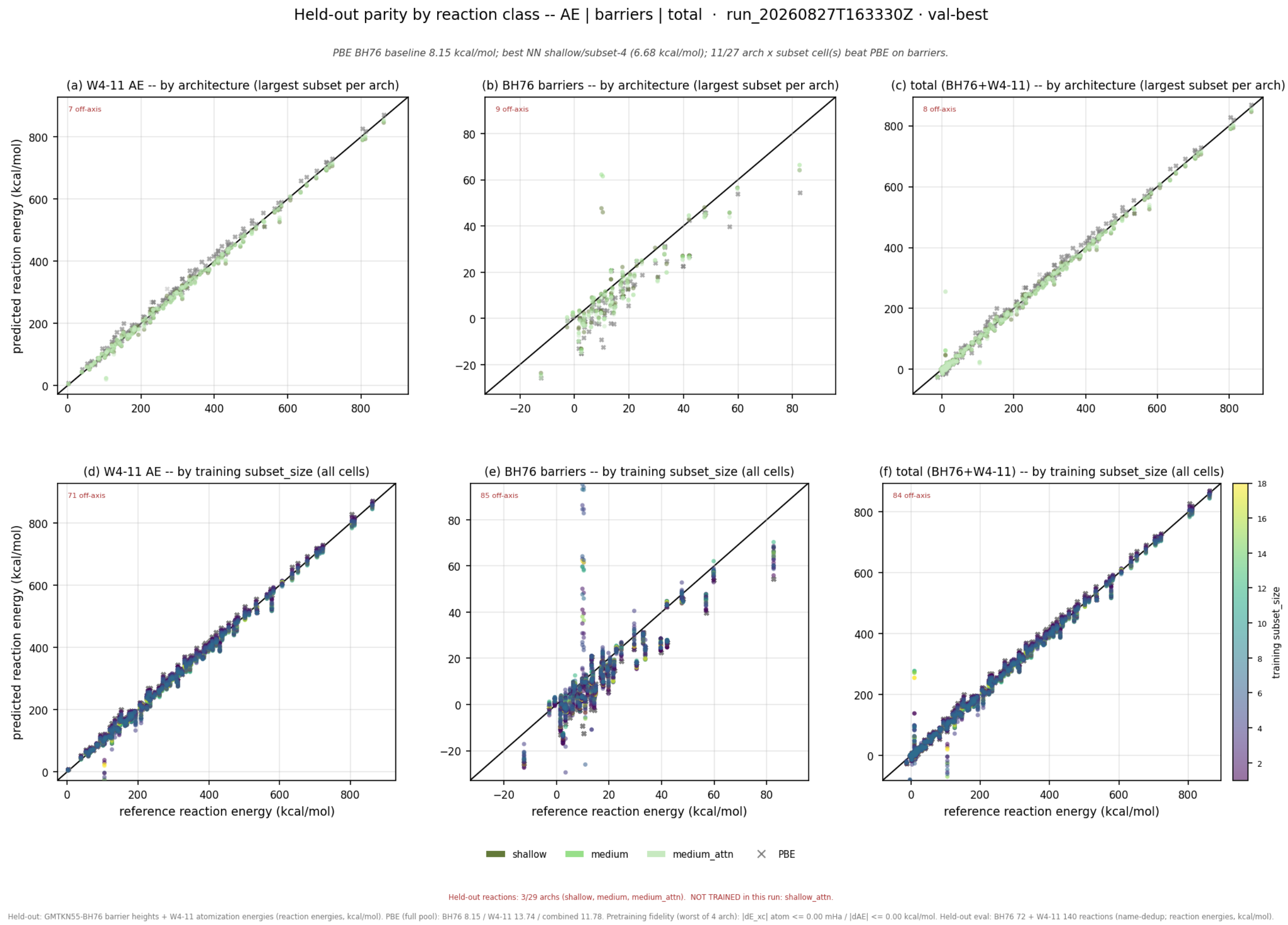

Figure 19: Held-out parity split by reaction class: W4-11 atomization, BH76 barriers, and the combined total. Top row coloured by architecture at each one's largest cell, with PBE as grey crosses; bottom row every cell coloured by training subset size.

Read the axis ranges first, since they are why the classes must be separated: on `medium` ss=18 the W4-11 references span 39.0 to 1007.9 kcal/mol while the BH76 references span −12.3 to 104.8 (from `per_reaction.json` of spec_0009), so on shared axes the barrier column would be a point. Read each column against its own diagonal. The W4-11 column shows the network tracking atomization energies over three decades with visible improvement on PBE. The BH76 column shows the scatter that the MAE table quantifies: on the same cell the network's barrier predictions run −24.5 to 87.8 kcal/mol against references of −12.3 to 104.8, i.e. it both over- and under-shoots. **Conclusion:** the anchored networks have learned atomization energetics and have not learned barrier heights; the combined column is dominated by W4-11 simply because that pool contributes about twice as many reactions per slice.

9.3 The c2 reference-branch incident and its repair

One species in the pool required a repair between the 25-cell and 27-cell states, and the episode is worth recording because it is a class of defect that internal consistency cannot catch.

C2 is a multireference dimer, and its PBE SCF at 6-311++G(3df,2pd) / grid 3 under the 3×10^{-5} orientation lock never converges in DIIS – 100 cycles oscillating between two SCF solutions with a trajectory spread of 1.204×10^{-1} Ha. The trajectory-best rescue introduced for the Li atom then flipped C2's reference from −75.8167407121 Ha (internally stable) to the internally unstable solution 50.10 kcal/mol above it, converged at −75.7368945310 Ha and stamped into the affected evaluations as −75.7368945258 Ha (the two differ at the eleventh decimal, the stamped value being the one the patch tool reads back as `from_E_pbe`), because PySCF's second-order SCF ingests a starting density by aufbau re-occupation of its Fock matrix and C2's ground solution is **non-aufbau in its own Fock** (an occupied orbital 2.35×10^{-4} Ha above a virtual). Any density start near the crossing therefore lands on either branch draw-dependently; four of ten local draws of the pre-fix code stamped the higher branch, as did seven pulled evaluations of `run_20260827T163330Z` (HISTORY 2026-08-31). The cross-spec reference guard, established on a different 24 mHa incident (HISTORY Phase 38), excluded c2 from the pooled figure baselines while the spread stood.

The mechanism is closed at its source: the DIIS recorder now also keeps the trajectory's lowest-energy point as an orbital pair, and any converged solution sitting more than 10^{-4} Ha above that minimum is rerun from the exact determinant – immune to re-occupation – with a standing excess refused rather than stamped (`data._-converge_reference_scf`). The tolerance is anchored to four measured same-basin excesses (-2.97×10^{-7} Ha for Li/SCAN, -4.09×10^{-6} for C2/PBE, $+8.26 \times 10^{-7}$ for the O/PBE rescue and $+8.38 \times 10^{-6}$ for the S/SCAN DIIS endpoint over its own trajectory minimum), sitting about 12 times above the largest of those and nearly three decades below the $+7.984 \times 10^{-2}$ Ha wrong-branch signal (`data.py:414-429`).

The already-written evaluations were repaired surgically rather than re-run: `hpcjobs/reeval_c2_patch.py` audits every channel against the two measured branch anchors, recomputes the C2 reference once through the branch-checked rescue, recomputes the C2-derived network quantities per PBE-seeded channel, and rewrites only the C2 entries of `per_molecule.json`, the `w411_c2_atomization` row of `per_reaction.json`, the containing slice rows of `test_set.csv` and a `reference_patch` stamp in `eval_metadata.json`, with the CSV and JSON reconstruction proven byte-identical on all 99 channels that existed when the tool ran before any patch logic executed (HISTORY 2026-09-01). Those 99 were the 25 then-complete specs' $25 \times 4 = 100$ channels less one: `spec_0026` had not yet written a cold-start artifact. The run now carries 108, which is 27×4 – the two `shallow` cells of Section 9.2 contributing 8 channels and `spec_0026`'s completed cold-start the ninth. No training subset ever contained c2 – the reaction `w411_c2_atomization` sits in the strict held-out slice of every completed cell with an empty `in_sample_overlap` – so no trained checkpoint was affected.

The state as pulled, verified by execution over the 27 completed specs: the seven repaired specs (0019, 0020 and 0022-0026) carry a `reference_patch` stamp on three channels each – `eval_holdout`, `eval_holdout_val_best` and `eval_holdout_coldstart` – recording `from_E_pbe` = −75.7368945258 Ha and `to_E_pbe` = −75.8167407121 Ha. On those three channels the C2 atomization's PBE error now reads −3.5457 kcal/mol in all 27 cells, against −53.6499 kcal/mol before, so the cross-spec reference guard is silent and the species has rejoined every pooled PBE baseline. **The `eval_holdout_best` channel is not repaired**: 7 of its 27 cells still carry −53.6499, pending a checkpoint fetch that must follow the push of the patched channels back to the cluster. That channel is therefore excluded from every table in this document – which costs nothing, since it is the channel the figure suite stopped plotting for independent reasons (Section 1.4).

9.4 The corrected v4gga validation-best record, per architecture (54 cells)

Within-cell beats over the same three slices (recomputed from `merged_v4_arms/checkpoints/spec_*/eval_-holdout_val_best/test_set.csv`; cell-matched PBE anchors: W4-11 13.60–14.00, BH76 7.42, combined 11.59–11.96):

arch	cells	W4-11	BH76	combined
deep_3x16	11	11/11	11/11	11/11
deep_attn_3x16	11	9/11	3/11	9/11
deep_cusp_3x16	11	5/11	6/11	5/11
deep_rung35_3x16	11	2/11	7/11	2/11
deep_rung35_attn_3x16	3	1/3	0/3	1/3
deep_rung35ms_3x16	7	0/7	0/7	0/7
total	54	28/54	27/54	28/54

Best cells: W4-11 4.51 (deep_attn, ss=26), BH76 4.14 (deep_3x16, ss=2), combined 4.79 (deep_attn, ss=26). The unanchored deep_3x16 curve beats PBE on *every* slice at *every* subset size – BH76 4.14–6.52, W4-11 7.12–11.41, combined 6.84–9.74.

The merged directory has since acquired 7 `deep_mgga_3x16` cells (3/7 on W4-11, 4/7 on BH76, 3/7 combined), taking it to 61 cells; those are a meta-GGA record and are excluded from the GGA table above and from every GGA comparison in this document. Note also that c2 is *absent* from the strict slice of 54 of the merged cells and present with a PBE error of −3.5638 kcal/mol in the remaining 7, so the v4 record is not affected by the incident of Section 9.3.

9.5 Reading

At matched capacity (medium pair against deep_3x16 pair), the anchored cells reproduce the unanchored campaigns' W4-11 gains – and do so uniformly (27 of 27 against 28 of 54; the unanchored total is diluted by the retired-fidelity descriptor arms, whose pretraining never delivered their parent, Section 4.2). On BH76 the picture inverts: the unanchored deep_3x16 beats PBE in all 11 cells (best 4.14) where the anchored cells beat it in 9 of 27 (best 6.46), and the signed decomposition (Section 8.3) locates the difference in the parent's barrier bias, which the unanchored large-s correlation freedom removed and the anchored parameterization largely retains. Whether that is the anchor's price or a removable training artifact is exactly what the queued anchored deep_3x16 group measures (Section 12). The comparison carries the coverage caveats stated: 27 of 44 G1 cells, no `shallow_attn` cell at all, one run per cell, and slices cell-matched within each campaign but not identical across campaigns.

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10. Energy and density together: the DFS-unit comparison

Everything above scores energies. The DFS Letter's central claim, however, is about the *density*: that a functional trained through a differentiable SCF improves the self-consistent density and not only the energy. The figure suite therefore carries a second family of views in which energy and density errors are combined on one axis, and this section reads them across generations.

10.1 The unit convention

Two quantities and one conversion define it.

The density error. The Letter's per-electron L^1 density error (its Eq. 20) is implemented in `xcquinox/alec/evaluation.py:187-209` as

$$\varepsilon_{|n|} = \frac{\sum_i w_i \left|\rho_i - \rho_i^{\mathrm{ref}}\right|}{\sum_i w_i \, \rho_i^{\mathrm{ref}}},$$

the numerator and the electron count N_e taken on the *same* quadrature so that grid truncation partially cancels, and N_e formed from the **reference** density so the measure is charge-correct for ions (the vendored dpyscf instead counts neutral-atom Z). It is dimensionless – electrons per electron – and is emitted per species as `density_eps_l1` alongside a model-free PBE twin `density_eps_l1_pbe`. It is *not* the same quantity as the suite's other density column, `density_rmse`, which is the grid-weight-averaged root mean square $\sqrt{\sum_i w_i(\Delta\rho_i)^2/\sum_i w_i}$ (`evaluation.py:283-291`); the two differ by about a factor of 40 in magnitude on this basis and grid, and mixing them is a units error. The equation number is the repository's own, verified against the Letter's PDF when the figure notation was fixed and stated as such (HISTORY 2026-08-03); the PDF is not in the repository, so the provenance of the equation numbering is the repo record.

The combined metric. The Letter's Eq. 21 combines an energy error E (WTMAD-2, kcal/mol) and a density error D by a harmonic mean,

$$\mathcal{ED} = \frac{2}{\dfrac{1}{E} + \dfrac{1}{\gamma D}},$$

so that a functional must be good on both to score well.

The conversion slope. γ carries units of kcal/mol per unit of $\varepsilon_{|n|}$. The Letter fixes it as the zero-intercept regression of WTMAD-2 on $\varepsilon_{|n|}$ across six nonempirical functionals (PW91, PBE, TPSS, revTPSS, SCAN, PBE0; its Fig. 3), obtaining $\gamma = 1084.87$ kcal/mol on its own diet-GMTKN55 axes. That published value is the only γ hardcoded in the repository (`make_ablation_arch_figure.py:4375-4381`, `_DFS_GAMMA_KCAL`). The same procedure re-run on this project's basis, grid, reference set and reaction list yields an **own-axes** slope of

$\gamma = 1158.336985911894$ kcal/mol, fitted by $\gamma = \sum \varepsilon W / \sum \varepsilon^2$ over the same six functionals (`make_ablation_arch_figure.gamma_zero_intercept` and `nonempirical_gamma`, `:4394-4404` and `:4506-4583`; the calibration data from `notebooks/analysis/precompute_nonempirical_pool.py`). That value was independently re-derived from the calibration JSON and the canonical reaction list and agreed with the pipeline to one unit in the last place, over all 216 reactions computable for all six functionals (HISTORY 2026-08-02).

Which slope a given figure actually plots is decided at render time – the own-axes fit when its calibration cache resolves, the published slope otherwise – and each panel stamps the value and source it used (`make_ablation_arch_figure.py:8153-8158`). **Verified by execution over all six DFS-units 3x3 CSVs read for this section: only the v3 figure sets carry the own-axes legs at 1158.336985911894; the v4gga, v4, v5, v6 G1 and merged-v4-GGA sets embedded below all carry** $\gamma = 1084.87$, the Letter's published value, because the calibration cache does not resolve for those runs. Every $\mathcal{ED}$ number quoted in this section is therefore at the published slope, and cross-generation comparisons of $\mathcal{ED}$ below are internally consistent for that reason.

One further distinction separates the two 3x3 variants. In the plain-units figure, γ is *self-calibrated per channel* as $E_{\mathrm{PBE}}/D_{\mathrm{PBE}}$ from that channel's own anchors, so $\mathcal{ED}$ of PBE equals E_{PBE} by construction and the three columns cannot be compared with one another. In the DFS-units figure one external γ is shared by all three columns, so $\mathcal{ED}_{|n|}$ *does* compare across columns and $\mathcal{ED}_{|n|}$ of PBE does not equal E_{PBE} (`make_ablation_arch_figure.py:6203-6224`, the two caveat strings).

10.2 The anchored generation

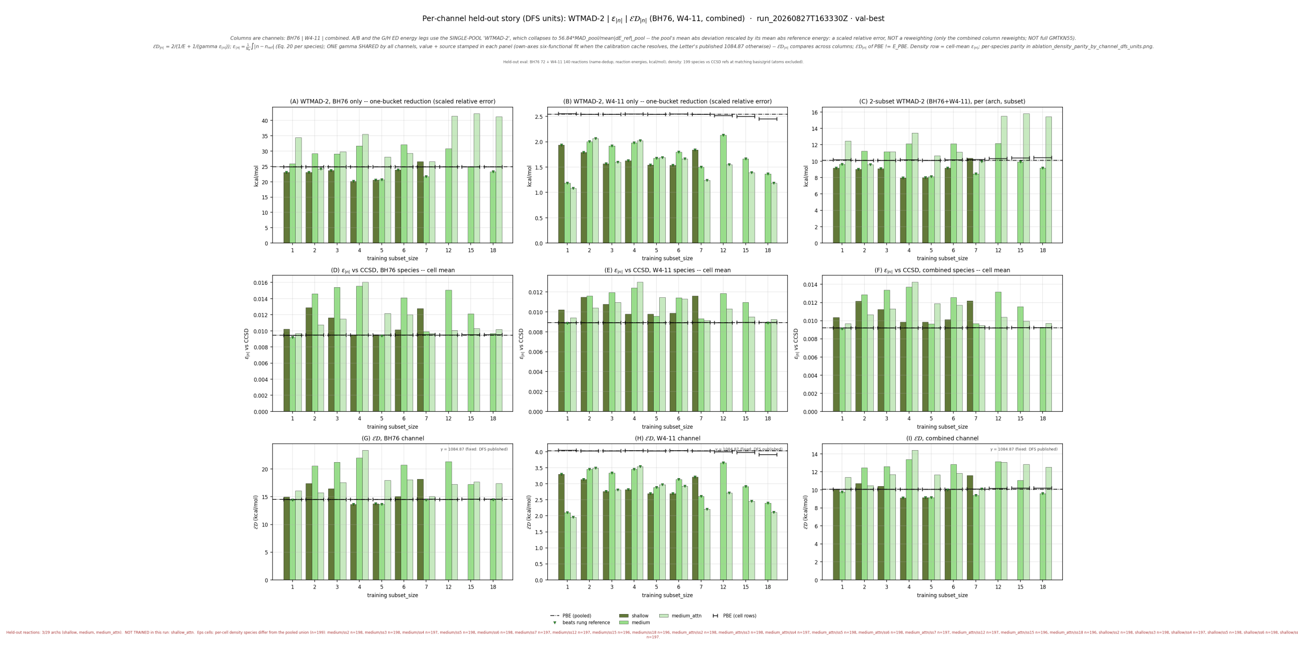

Figure 20: Per-channel held-out story in DFS units, anchored v6 G1. Columns: BH76, W4-11, combined. Row 1: WTMAD-2 per cell (kcal/mol). Row 2: cell-mean $\varepsilon_{|n|}$ against CCSD. Row 3: the combined metric $\mathcal{ED}_{|n|}$ at one shared γ. PBE dashed in every panel.

Read down a column, not across a row: row 1 is what the network did to the energy, row 2 what it did to the density, row 3 the combination. From `ablation_density_energy_3x3_dfs_units.csv` (27 cells per column, $\gamma = 1084.87$):

channel	WTMAD-2, NN / PBE	$\varepsilon_{\|n\|}$, NN / PBE	$\mathcal{ED}_{\|n\|}$, NN / PBE	energy beats	density beats	$\mathcal{ED}$ beats
BH76	20.19–42.26 / 24.85	0.00920–0.01607 / 0.00946	13.63–23.39 / 14.53	10/27	2/27	6/27
W4-11	1.08–2.13 / 2.55	0.00885–0.01302 / 0.00893	1.95–3.66 / 4.03	27/27	2/27	27/27
combined	7.96–15.82 / 10.12	0.00912–0.01427 / 0.00921	9.12–14.39 / 10.06	13/27	1/27	8/27

channel	WTMAD-2, NN / PBE	$\varepsilon_{\|n\|}$, NN / PBE	$\mathcal{ED}_{\|n\|}$, NN / PBE	energy beats	density beats	$\mathcal{ED}$ beats

The finding is in row 2. **The anchored cells improve the energy without improving the density**: on the combined channel exactly one of 27 cells has a smaller per-electron density error than PBE, and the best cell's $\varepsilon_{|n|}$ (0.00912) is 1 percent below PBE's 0.00921. The same asymmetry on W4-11, where the best-$\varepsilon_{|n|}$ cell is `medium` at ss=1: its density error is 0.86 percent better than PBE's (0.008855 against 0.008932) while its WTMAD-2 is **53.56 percent** better (1.1821 against 2.5455), or 52.25 percent on the plain mean absolute error of Section 9.2. The energy is beaten in every W4-11 cell and the density in two. **Conclusion:** at this coverage the anchored networks are energy-fitted functionals whose self-consistent densities are indistinguishable from – and usually slightly worse than – their parent's. That is precisely the failure mode the DFS protocol's density-weighted loss exists to prevent, and it is a live finding, not a settled one: the loss carries the Letter's density weight of 20 against reaction weight 1 (`LOSS_PRIMER.md:42-56`, verified to 2.2×10^{-16} over all 1400 optimizer updates), so the density channel is not being ignored by the objective. Note also that WTMAD-2 here is the one-bucket reduction on BH76 and W4-11 separately and the genuine two-subset form on the combined column, so the row-1 numbers are not the mean absolute errors of Section 9.2 and should not be read against them.

One presentational difference from Section 9 is worth stating, because the two "beats PBE" counts are not the same test. Section 9 compares each cell against **its own slice's** PBE anchor, so the anchor varies by cell; the `beats_pbe` column of these CSVs compares against a single **pooled** PBE anchor per channel, which is why the PBE column above is one number rather than a range. The two agree on the qualitative picture and need not agree cell for cell.

10.3 The unanchored generations

The same three rows for the unanchored GGA arm, per-arm slice. From that directory's `ablation_density_-energy_3x3_dfs_units.csv`, at the same $\gamma = 1084.87$: BH76 WTMAD-2 spans 11.60–543.47 against PBE's 21.23, $\varepsilon_{|n|}$ spans 0.00759–0.04746 against 0.00943, and $\mathcal{ED}_{|n|}$ spans 11.60–94.07 against 13.81; the beat counts are 27/54 on energy, **27/54 on density**, and 31/54 on the combined metric. The W4-11 column reads 27/54, 15/54 and 28/54 on the same three legs, the combined column 25/54, 18/54 and 27/54.

Two things separate this from the anchored panel. The **range** is enormous – three of the retired-fidelity descriptor architectures produce WTMAD-2 values in the hundreds and density errors five times PBE's, which is what a network that started 25.7 to 56.1 kcal/mol from its parent on its worst system and then diverged looks like. But the **density beat rate is an order of magnitude higher**: 27 of 54 BH76 cells improve the per-electron density error where 2 of 27 anchored cells do, and the best unanchored density error, 0.00759, is 20 percent below PBE's where the best anchored one is 3 percent below. **Conclusion:** the unanchored parameterization moved the density and the anchored one has not. Read together with Section 8.2 this is consistent: the large-gradient correlation freedom the anchor suppresses is exactly the region that distinguishes one self-consistent density from another, and the anchored networks have been buying their energy improvements in the small-gradient region where the pre-image leaves them trainable.

This third panel is embedded because it is not a redundant view of the previous one: it is the same architectures scored on a **different slice** – the cross-arm merged slice, which removes the union of both arms' validation reactions rather than each arm's own, and which drops the diverged rung-3.5-multishell cells. It is the cleaner statement of what the unanchored protocol achieved when it worked. From its CSV: BH76 WTMAD-2 8.92–41.81 against PBE's 16.48, $\varepsilon_{|n|}$ 0.00681–0.01338 against 0.00903, $\mathcal{ED}_{|n|}$ 9.64–19.04 against 12.29, with **43 of 47 cells beating PBE on the combined metric, 41 of 47 on energy and 30 of 47 on density**. No single architecture wins every leg: the best WTMAD-2 is `deep_3x16` at ss=6 (8.92084), the best $\varepsilon_{|n|}$ is `deep_cusp_3x16` at ss=26 (0.006805), and the best $\mathcal{ED}_{|n|}$ is `deep_attn_3x16` at ss=18 (9.63922) – the descriptor architecture takes the density leg while the descriptor-free one takes the energy leg. **Conclusion:** on a slice where the failed architectures do not dominate the range, the unanchored protocol improved the density in about two thirds of its BH76 cells – a genuine density result of the kind the Letter reports, and the specific thing the anchored generation has not yet reproduced. It also carries the caveat that makes it not a clean comparison target: those same cells' pretrained networks were 25.7 to 56.1 kcal/mol from their parents on their worst systems, so their density improvement is not attributable to a faithful starting point.

Figure 21: Per-channel held-out story in DFS units, unanchored v4gga, six architectures at 54 cells.

Figure 22: Per-channel held-out story in DFS units, the merged v4 GGA record on the union-removed comparison slice, 47 cells.

10.4 The plain-units view, and why one is enough

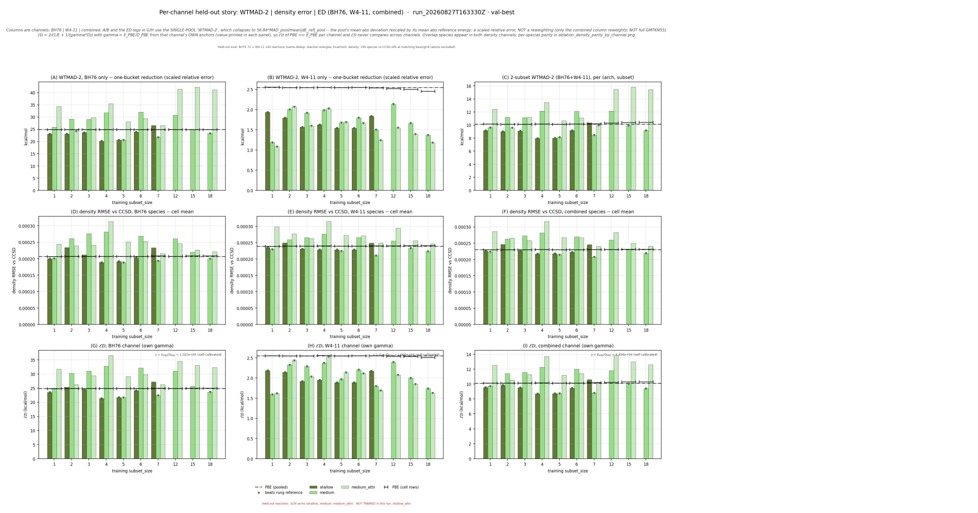

Figure 23: Per-channel held-out story in plain units, anchored v6 G1: row 2 is the grid-weight-averaged density RMSE and row 3 uses a per-channel self-calibrated γ.

The plain-units twin is the same data under a different density measure and a different conversion. Row 1 is byte-for-byte the DFS-units row 1. Row 2 plots `density_rmse` instead of $\varepsilon_{|n|}$: from `ablation_density_energy_-3x3.csv` the combined-channel values span 2.075×10^{-4} to 3.172×10^{-4} against PBE's 2.298×10^{-4}, where the same cells' $\varepsilon_{|n|}$ span 0.00912 to 0.01427 – the two measures differ by a factor of about 40 and, being an L^2 and an L^1 measure on different normalizations, they are not interconvertible by a constant. Row 3 uses the self-calibrated slope, which the CSV records as $\gamma = 120154.3$ (BH76), 10656.7 (W4-11) and 44039.3 (combined) kcal/mol per unit RMSE – by construction $E_{\mathrm{PBE}}/D_{\mathrm{PBE}}$ per channel, which is why PBE's $\mathcal{ED}$ equals its WTMAD-2 exactly in every column (24.85, 2.55, 10.12) and why these columns must not be compared with one another.

Conclusion, and the redundancy judgment: the plain-units view reproduces the qualitative finding of Section 10.2 – the anchored cells improve energy and not density – under an independent density norm, which is the reason to show it once. It adds nothing beyond that, and its self-calibrated γ makes it strictly less useful for cross-channel and cross-generation comparison, so the plain-units twins of the other generations, the `_logy` variants of all of them, and the `ablation_density_energy_overview*`, `ablation_ed_decomposition*` and `ablation_density_parity_by_channel*` families (which re-decompose the same three quantities per species and per architecture) are not reproduced here.

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11. The generation comparison

Every cell of the following table is sourced in the section named beside it.

	v4 / v4gga	v5 / v5mgga2	v6
Pretraining objective	integration-weighted point-wise F residual, 2500 steps; no energy term, no validation, no gate (Sec. 4.1)	unchanged from v4 – the shared pretrain data file could not regenerate, so the pretrained pairs are the v4 objects (Sec. 5)	same residual plus an optional per-system parent-energy term, run at weight 0.0 under the anchor because both terms are zero at initialization; 20% seeded validation with patience, best-validation network written (Sec. 6.3)
SCF seeding	converged PBE for every rung	converged PBE for GGA, converged SCAN for meta-GGA (`seed_xc: auto`); a fourth cold-start channel added (Sec. 5, Sec. 1.4)	as v5

	v4 / v4gga	v5 / v5mgga2	v6
Anchor	none: $F = 1 + L(T(g))$, start at the LDA limit $F \equiv 1$ (Sec. 2)	none	$F = 1 + L(z_{\text{parent}} + T(g))$; the parent returned exactly at $T = 0$, to one unit in the last place at $\Lambda = 1.174$ and bitwise at 1.804 and 2.0 (Sec. 6.1)
Coordinates	legacy: s through $\{1 - e^{-x^2}\} \ln(x + 1)$, C-net density feature r_s through the same s-style transform – a documented deviation from DFS Eq. 7 (Sec. 3.7)	as v4	DFS: x_s (Eq. 9), $x_0 = \ln(\rho^{1/3} + 10^{-5})$ (Eq. 7), x_1 (Eq. 4), x_α (Eq. 10) (Sec. 3.7)
Open-shell footing	total-density descriptors passed into both spin channels – the meta-GGA spin-scaling defect (Sec. 4.3)	unrepaired	symmetric doubled density $\text{diag}(P_\sigma, P_\sigma)$ everywhere; libxc spin-polarized SCAN exchange reproduced to 1.8×10^{-15} Ha on O and OH (Sec. 6.2)
Acceptance gate	none	none	per-architecture fidelity certificate: PASS requires max atomic $\lvert\Delta E_{xc}\rvert \leq 1.0$ mHa and max$\lvert\Delta AE\rvert \leq 1.0$ kcal/mol on 38–39 systems through the production energy path; enforced unconditionally by every record layer (Sec. 6.4)
First-step pretrain loss	1.591407×10^{-2}, identical across all six GGA architectures (the v4gga run's own `pretrain/*/losses_x.npy`); the unanchored G1 control submission reads 0.008–0.012 (Sec. 6.4)	as v4	2.72×10^{-32} (PBE parent); $3.02–4.31 \times 10^{-14}$ (SCAN parent, the `_ALPHA_MAX` ceiling, Sec. 6.5)
Handoff fidelity, curve metric	max$\lvert\Delta F_x\rvert$ 0.039–0.090 (GGA); meta-GGA max$\lvert\Delta F_c\rvert$ up to 0.49 (Sec. 4.2, Sec. 5)	identical to v4 by construction; the stacking arm reaches 0.52 (Sec. 5)	8.7×10^{-7}–9.2×10^{-6} in F_x (G1); 8.1×10^{-7}–1.3×10^{-5} over both channels (meta-GGA) (Sec. 6.6)
Handoff fidelity, energy	atomization offsets −2.3 to −56.1 kcal/mol on H2O/N2/CH4; worst-system offset per descriptor-carrying architecture 25.7–56.1 kcal/mol against 4.1–4.2 for the descriptor-free controls, individual systems overlapping (Sec. 4.2)	unmeasured until 2026-08-20, then the same (Sec. 5)	certificates PASS at 7.2×10^{-4}–5.15×10^{-3} mHa atomic and 1.9×10^{-3}–2.5×10^{-3} kcal/mol on atomization (Sec. 6.4)
Held-out headline (GGA, validation-best, within-cell)	W4-11 28/54, BH76 27/54, combined 28/54; best cells 4.51 / 4.14 / 4.79 kcal/mol (Sec. 9.4)	no GGA cells (its GGA rows are the v4 rows)	W4-11 27/27, BH76 9/27, combined 26/27 at 27 of 44 cells; best cells 5.99 / 6.46 / 7.00 kcal/mol (Sec. 9.2)
Held-out density (DFS units, combined channel)	$\varepsilon_{\lvert n \rvert}$ beats PBE in 18/54 on the per-arm slice and 18/47 on the merged slice (both from the `ablation_density_energy_3x3_-dfs_units.csv` of the respective directory); BH76 30/47 on the merged slice (Sec. 10.3)	–	1/27 (Sec. 10.2)
Status of the record	retired for every descriptor-carrying architecture; the descriptor-free `deep_3x16` and `deep_attn_3x16` rows stand (Sec. 4.3)	meta-GGA record retired as a quantitative result; v5mgga2 never trained (Sec. 5)	live

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12. Open questions the next results answer

1. **The controlled anchored-vs-unanchored test.** The G2a core trio (deep_3x16, deep_attn_3x16, deep_cusp_3x16 under the full v6 protocol with `parent_anchor: true`) is queued behind the draining G1 group (HISTORY 2026-08-30, the trio split; HISTORY 2026-08-31). It meets the strongest unanchored record at the registry identity itself: the G1 medium pair already realizes the deep_3x16 capacity with both differing flags inert for that descriptor-free pair (Section 9.1), so the trio's deep_3x16 and deep_attn cells test the anchored result's reproducibility at the registry names themselves, and its deep_cusp cells extend the comparison to a descriptor form G1 does not carry; the two group files share an identical subset axis including ss=26 (`dfs_step7.dfs6311_grid3_v6g1_size.yaml` line 129, `...v6g2a_families_core.yaml` line 122), so G1's ss=26 cells are pending rather than absent (item 4). The BH76 signed bias (Section 8.3) is the discriminating observable; the G1 spread (−7.75 at medium/ss=12 against −0.81 at medium_attn/ss=12)

says the outcome is not foreclosed.

2. **The density question.** The anchored G1 cells improve energies almost everywhere and the per-electron density error almost nowhere – 1 of 27 combined-channel cells against 18 of 47 for the unanchored merged record (Section 10). Two candidate explanations are separable by the queued groups: that the pre-image suppression of large-s correlation (Section 8.2) removes the freedom a density improvement needs, in which case the anchored deep_cusp and rung-3.5 cells will show the same pattern; or that the unanchored density improvements were an artifact of networks that started far from their parent and had large corrections to build, in which case they should not survive the anchored re-run of the same architectures.
3. **The meta-GGA trained factors.** The five anchored meta-GGA family architectures hold PASS certificates at production identity and pretrained curves within 8.1×10^{-7}–1.3×10^{-5} of SCAN (Sections 6.4-6.6); their training cells are the pending half of `figures_dfs_step7_dfs6311_grid3_v6g2_families_mgga`. They answer whether the anchored fine-tune preserves SCAN-level held-out accuracy where the v5 SCAN-seeded (but mis-footed, unanchored) cells only matched it at subset sizes 2–5 (Section 5), and whether the correlation-collapse mechanism of Section 8.2 repeats against the SCAN parent – whose correlation, unlike PBE's, retains a finite floor at large s (Section 2), so the suppression should be materially weaker.
4. **G1 completion.** 17 G1 cells remain (all 11 shallow_attn cells, the four remaining shallow sizes, and the ss=26 cells of medium and medium_attn – the medium ss=26 cell needs the open NaN-gradient defect closed, HISTORY 2026-08-31), after which the medium-pair-vs-anchored-deep-pair comparison of item 1 – registry names differing in three fields, all inert under the v6 model block for the descriptor-free pair (Section 9.1) – can be read at full coverage. Until then the size ladder's attention comparison is one-sided: no `shallow_attn` cell exists. Also pending is the checkpoint fetch that lets the `eval_holdout_best` channel be repaired for the seven c2-affected specs (Section 9.3).

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13. Sources

Primary literature.

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Figure-generating code. `notebooks/analysis/report_equation_figures.py` (the governing-equation figures, drawn by calling the repository's own functions), `notebooks/analysis/pretrain_fx_fc.py` and `trained_fx_fc.py` (the per-generation enhancement-factor sets), `notebooks/analysis/anchored_vs_unanchored_fx_fc.py` (the cross-generation 2x2), and `notebooks/analysis/make_ablation_arch_figure.py` (the `ablation_*` suite). The equation-figure and enhancement-factor modules write a long-form CSV for every figure they render; the `ablation_*` suite does so only for some, and the three `ablation_*` figures embedded here have none. Every number quoted in a caption above was read either from that figure's own CSV or, where none exists, from the per-spec artifacts named in the caption.